

《线性代数》

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重难知识点与常考试题之经典汇总

《线性代数》

第一部分 知识点整理

一 行列式

1, n 行列式共有 n^2 个元素, 展开后有 $n!$ 项, 可分解为 2^n 行列式;

2, 代数余子式的性质:

①、 A_{ij} 和 a_{ij} 的大小无关; ②、某行(列)的元素乘以其它行(列)元素的代数余子式为 0;

③、某行(列)的元素乘以该行(列)元素的代数余子式为 $|A|$;

3, 代数余子式和余子式的关系:

$$M_{ij} = (-1)^{i+j} A_{ij} \quad A_{ij} = (-1)^{i+j} M_{ij}$$

4, 设 n 行列式 D :

将 D 上、下翻转或左右翻转, 所得行列式为 D_1 , 则 $D_1 = (-1)^{\frac{n(n-1)}{2}} D$;

将 D 顺时针或逆时针旋转 90° , 所得行列式为 D_2 , 则 $D_2 = (-1)^{\frac{n(n-1)}{2}} D$;

将 D 主对角线翻转后(转置), 所得行列式为 D_3 , 则 $D_3 = D$;

将 D 主副角线翻转后, 所得行列式为 D_4 , 则 $D_4 = D$;

5, 行列式的重要公式:

①、主对角行列式: 主对角元素的乘积; ②、副对角行列式: 副对角元素的乘积 $\times (-1)^{\frac{n(n-1)}{2}}$;

③、上、下三角行列式 ($|\begin{smallmatrix} \blacktriangledown \end{smallmatrix}| = |\begin{smallmatrix} \blacktriangle \end{smallmatrix}|$): 主对角元素的乘积; ④、 $|\begin{smallmatrix} \blacktriangledown \end{smallmatrix}|$ 和 $|\begin{smallmatrix} \blacktriangle \end{smallmatrix}|$: 副对角元素的乘积 $\times (-1)^{\frac{n(n-1)}{2}}$;

⑤、拉普拉斯展开式:

$$\begin{vmatrix} A & O \\ C & B \end{vmatrix} = \begin{vmatrix} A & C \\ O & B \end{vmatrix} = |A||B|, \quad \begin{vmatrix} C & A \\ B & O \end{vmatrix} = \begin{vmatrix} O & A \\ B & C \end{vmatrix} = (-1)^{m \cdot n} |A||B| \quad \text{⑥、范德蒙行列式: 大指标减小指标的连乘积; ⑦、特征值;}$$

6, 对于 n 阶行列式 $|A|$, 恒有: $|\lambda E - A| = \lambda^n + \sum_{k=1}^n (-1)^k S_k \lambda^{n-k}$, 其中 S_k 为 k 阶主子式;

7, 证明 $|A| = 0$ 的方法:

①、 $|A| = -|A|$; ②、反证法; ③、构造齐次方程组 $Ax = 0$, 证明其有非零解; ④、利用秩, 证明 $r(A) < n$; ⑤、证明 0 是其特征值;

二 矩阵

1, A 是 n 阶可逆矩阵:

$\Leftrightarrow |A| \neq 0$ (是非奇异矩阵); $\Leftrightarrow r(A) = n$ (是满秩矩阵) $\Leftrightarrow A$ 的行(列)向量组线性无关;

$\Leftrightarrow$ 齐次方程组 $Ax = 0$ 有非零解; $\Leftrightarrow \forall b \in R^n, Ax = b$ 总有唯一解; $\Leftrightarrow A$ 与 E 等价;

$\Leftrightarrow A$ 可表示成若干个初等矩阵的乘积; $\Leftrightarrow A$ 的特征值全不为 0; $\Leftrightarrow A^T A$ 是正定矩阵;

$\Leftrightarrow A$ 的行(列)向量组是 R^n 的一组基; $\Leftrightarrow A$ 是 R^n 中某两组基的过渡矩阵;

2, 对于 n 阶矩阵 A : $AA^* = A^*A = |A|E$ 无条件恒成立;

$$(A^{-1})^* = (A^*)^{-1} \quad (A^{-1})^T = (A^T)^{-1} \quad (A^*)^T = (A^T)^*$$

$$(AB)^T = B^T A^T \quad (AB)^* = B^* A^* \quad (AB)^{-1} = B^{-1} A^{-1}$$

4, 矩阵是表格, 推导符号为波浪号或箭头; 行列式是数值, 可求代数和;

5, 关于分块矩阵的重要结论, 其中均 A 、 B 可逆:

$$\text{若 } A = \begin{pmatrix} A_1 & & \\ & A_2 & \\ & & \ddots \\ & & & A_s \end{pmatrix}, \text{ 则:}$$

$$\text{I、 } |A| = |A_1||A_2|\cdots|A_s|;$$

$$\text{II、 } A^{-1} = \begin{pmatrix} A_1^{-1} & & \\ & A_2^{-1} & \\ & & \ddots \\ & & & A_s^{-1} \end{pmatrix};$$

$$\text{②、 } \begin{pmatrix} A & O \\ O & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & O \\ O & B^{-1} \end{pmatrix}; \quad (\text{主对角分块})$$

$$\text{③、 } \begin{pmatrix} O & A \\ B & O \end{pmatrix}^{-1} = \begin{pmatrix} O & B^{-1} \\ A^{-1} & O \end{pmatrix}; \quad (\text{副对角分块})$$

$$\text{④、 } \begin{pmatrix} A & C \\ O & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ O & B^{-1} \end{pmatrix}; \quad (\text{拉普拉斯})$$

$$\text{⑤、 } \begin{pmatrix} A & O \\ C & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & O \\ -B^{-1}CA^{-1} & B^{-1} \end{pmatrix}; \quad (\text{拉普拉斯})$$

三 矩阵的初等变换与线性方程组

1, 一个 $m \times n$ 矩阵 A , 总可经过初等变换化为标准形, 其标准形是唯一确定的: $F = \begin{pmatrix} E_r & O \\ O & O \end{pmatrix}_{m \times n};$

等价类: 所有与 A 等价的矩阵组成的一个集合, 称为一个等价类; 标准形为其形状最简单的矩阵;

对于同型矩阵 A, B , 若 $r(A) = r(B) \Leftrightarrow A \sim B$;

2, 行最简形矩阵:

①、只能通过初等行变换获得; ②、每行首个非 0 元素必须为 1; ③、每行首个非 0 元素所在列的其他元素必须为 0;

3, 初等行变换的应用: (初等列变换类似, 或转置后采用初等行变换)

①、若 $(A, E) \xrightarrow{r} (E, X)$, 则 A 可逆, 且 $X = A^{-1}$;

②、对矩阵 (A, B) 做初等行变化, 当 A 变为 E 时, B 就变成 $A^{-1}B$, 即: $(A, B) \xrightarrow{c} (E, A^{-1}B)$;

③、求解线性方程组: 对于 n 个未知数 n 个方程 $Ax = b$, 如果 $(A, b) \xrightarrow{r} (E, x)$, 则 A 可逆, 且 $x = A^{-1}b$;

4, 初等矩阵和对角矩阵的概念:

①、初等矩阵是行变换还是列变换, 由其位置决定: 左乘为初等行矩阵、右乘为初等列矩阵;

②、 $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$, 左乘矩阵 A , λ_i 乘 A 的各行元素; 右乘, λ_i 乘 A 的各列元素;

③、对调两行或两列, 符号 $E(i, j)$, 且 $E(i, j)^{-1} = E(i, j)$, 例如: $\begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix};$

④、倍乘某行或某列, 符号 $E(i(k))$, 且 $E(i(k))^{-1} = E(i(\frac{1}{k}))$, 例如: $\begin{pmatrix} 1 & & \\ & k & \\ & & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & & \\ & \frac{1}{k} & \\ & & 1 \end{pmatrix} (k \neq 0);$

⑤、倍加某行或某列, 符号 $E(ij(k))$, 且 $E(ij(k))^{-1} = E(ij(-k))$, 如: $\begin{pmatrix} 1 & & k \\ & 1 & \\ & & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & & -k \\ & 1 & \\ & & 1 \end{pmatrix} (k \neq 0);$

5, 矩阵秩的基本性质:

①、 $0 \leq r(A_{m \times n}) \leq \min(m, n)$; ②、 $r(A^T) = r(A)$; ③、若 $A \sim B$, 则 $r(A) = r(B)$;

④、若 P, Q 可逆, 则 $r(A) = r(PA) = r(AQ) = r(PAQ)$; (可逆矩阵不影响矩阵的秩)

⑤、 $\max(r(A), r(B)) \leq r(A, B) \leq r(A) + r(B)$; (※) ⑥、 $r(A+B) \leq r(A) + r(B)$; (※)

⑦、 $r(AB) \leq \min(r(A), r(B))$; (※) ⑧、如果 A 是 $m \times n$ 矩阵, B 是 $n \times s$ 矩阵, 且 $AB = 0$, 则: (※)

A 和 B 的列向量全部是齐次方程组 $AX = 0$ 解 (转置运算后的结论);

$\beta_1^T, \beta_2^T, \dots, \beta_m^T$ 构成 $m \times n$ 矩阵 $B = \begin{pmatrix} \beta_1^T \\ \beta_2^T \\ \vdots \\ \beta_m^T \end{pmatrix}$; 含有有限个向量的有序向量组与矩阵一一对应;

2, ①、向量组的线性相关, 无关 $\Leftrightarrow Ax = 0$ 有无非零解; (齐次线性方程组)

②、向量的线性表出 $\Leftrightarrow Ax = b$ 是否有解; (线性方程组)

③、向量组的相互线性表示 $\Leftrightarrow AX = B$ 是否有解; (矩阵方程)

3, 矩阵 $A_{m \times n}$ 与 $B_{l \times n}$ 行向量组等价的充分必要条件是: 齐次方程组 $Ax = 0$ 和 $Bx = 0$ 同解;

4, $r(A^T A) = r(A)$;

5, n 维向量线性相关的几何意义:

①、 α 线性相关 $\Leftrightarrow \alpha = 0$;

②、 α, β 线性相关 $\Leftrightarrow \alpha, \beta$ 坐标成比例或共线 (平行);

③、 α, β, γ 线性相关 $\Leftrightarrow \alpha, \beta, \gamma$ 共面;

6, 线性相关与无关的两套定理:

若 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关, 则 $\alpha_1, \alpha_2, \dots, \alpha_s, \alpha_{s+1}$ 必线性相关;

若 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性无关, 则 $\alpha_1, \alpha_2, \dots, \alpha_{s-1}$ 必线性无关; (向量的个数加加减减, 二者为对偶)

若 r 维向量组 A 的每个向量上添上 $n-r$ 个分量, 构成 n 维向量组 B :

若 A 线性无关, 则 B 也线性无关; 反之若 B 线性相关, 则 A 也线性相关; (向量组的维数加加减减)

简言之: 无关组延长后仍无关, 反之, 不确定;

7, 向量组 A (个数为 r) 能由向量组 B (个数为 s) 线性表示, 且 A 线性无关, 则 $r \leq s$

向量组 A 能由向量组 B 线性表示, 则 $r(A) \leq r(B)$;

向量组 A 能由向量组 B 线性表示

$\Leftrightarrow AX = B$ 有解;

$\Leftrightarrow r(A) = r(A, B)$ (P_{85} 定理 2)

向量组 A 能由向量组 B 等价 $\Leftrightarrow r(A) = r(B) = r(A, B)$

8, 方阵 A 可逆 $\Leftrightarrow$ 存在有限个初等矩阵 $P_1, P_2, \dots, P_l$, 使 $A = P_1 P_2 \dots P_l$;

①、矩阵行等价: $A \sim^r B \Leftrightarrow PA = B$ (左乘, P 可逆) $\Leftrightarrow Ax = 0$ 与 $Bx = 0$ 同解

②、矩阵列等价: $A \sim^c B \Leftrightarrow AQ = B$ (右乘, Q 可逆);

③、矩阵等价: $A \sim B \Leftrightarrow PAQ = B$ (P, Q 可逆);

9, 对于矩阵 $A_{m \times n}$ 与 $B_{l \times n}$:

①、若 A 与 B 行等价, 则 A 与 B 的行秩相等;

②、若 A 与 B 行等价, 则 $Ax = 0$ 与 $Bx = 0$ 同解, 且 A 与 B 的任何对应的列向量组具有相同的线性相关性;

③、矩阵的初等变换不改变矩阵的秩;

④、矩阵 A 的行秩等于列秩;

10, 若 $A_{m \times s} B_{s \times n} = C_{m \times n}$, 则:

①、 C 的列向量组能由 A 的列向量组线性表示, B 为系数矩阵;

②、 C 的行向量组能由 B 的行向量组线性表示, A^T 为系数矩阵; (转置)

11, 齐次方程组 $Bx = 0$ 的解一定是 $ABx = 0$ 的解, 考试中可以直接作为定理使用, 而无需证明;

①、 $ABx = 0$ 只有零解 $\Rightarrow Bx = 0$ 只有零解;

②、 $Bx = 0$ 有非零解 $\Rightarrow ABx = 0$ 一定存在非零解;

12, 设向量组 $B_{n \times r}: b_1, b_2, \dots, b_r$ 可由向量组 $A_{n \times s}: a_1, a_2, \dots, a_s$ 线性表示为: $(b_1, b_2, \dots, b_r) = (a_1, a_2, \dots, a_s)K$ ($B = AK$)

其中 K 为 $s \times r$, 且 A 线性无关, 则 B 组线性无关 $\Leftrightarrow r(K) = r$; (B 与 K 的列向量组具有相同线性相关性)

(必要性: $\because r = r(B) = r(AK) \leq r(K), r(K) \leq r, \therefore r(K) = r$; 充分性: 反证法)

注: 当 $r = s$ 时, K 为方阵, 可当作定理使用;

13, ①、对矩阵 $A_{m \times n}$, 存在 $Q_{n \times m}$, $AQ = E_m \Leftrightarrow r(A) = m$ 、 Q 的列向量线性无关;

②、对矩阵 $A_{m \times n}$, 存在 $P_{n \times m}$, $PA = E_n \Leftrightarrow r(A) = n$ 、 P 的行向量线性无关;

14, $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关

$\Leftrightarrow$ 存在一组不全为 0 的数 $k_1, k_2, \dots, k_s$, 使得 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$ 成立;

$$\Leftrightarrow (\alpha_1, \alpha_2, \dots, \alpha_s) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_s \end{pmatrix} = 0 \text{ 有非零解, 即 } Ax = 0 \text{ 有非零解};$$

$\Leftrightarrow r(\alpha_1, \alpha_2, \dots, \alpha_s) < s$, 系数矩阵的秩小于未知数的个数;

15, 设 $m \times n$ 的矩阵 A 的秩为 r , 则 n 元齐次线性方程组 $Ax = 0$ 的解集 S 的秩为: $r(S) = n - r$;

16, 若 η^* 为 $Ax = b$ 的一个解, $\xi_1, \xi_2, \dots, \xi_{n-r}$ 为 $Ax = 0$ 的一个基础解系, 则 $\eta^*, \xi_1, \xi_2, \dots, \xi_{n-r}$ 线性无关;

五 相似矩阵和二次型

1, 正交矩阵 $\Leftrightarrow A^T A = E$ 或 $A^{-1} = A^T$ (定义), 性质:

①、 A 的列向量都是单位向量, 且两两正交, 即

$$a_i^T a_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} (i, j = 1, 2, \dots, n);$$

②、若 A 为正交矩阵, 则 $A^{-1} = A^T$ 也为正交阵, 且 $|A| = \pm 1$;

③、若 A 、 B 正交阵, 则 AB 也是正交阵;

注意: 求解正交阵, 千万不要忘记施密特正交化和单位化;

2, 施密特正交化: $(a_1, a_2, \dots, a_r)$

$$b_1 = a_1;$$

$$b_2 = a_2 - \frac{[b_1, a_2]}{[b_1, b_1]} b_1$$

.....

$$b_r = a_r - \frac{[b_1, a_r]}{[b_1, b_1]} b_1 - \frac{[b_2, a_r]}{[b_2, b_2]} b_2 - \dots - \frac{[b_{r-1}, a_r]}{[b_{r-1}, b_{r-1}]} b_{r-1};$$

3, ①、 A 与 B 等价 $\Leftrightarrow A$ 经过初等变换得到 B ;

$$\Leftrightarrow PAQ = B, P, Q \text{ 可逆};$$

$$\Leftrightarrow r(A) = r(B), A, B \text{ 同型};$$

②、 A 与 B 合同 $\Leftrightarrow C^T AC = B$, 其中可逆;

$$\Leftrightarrow x^T Ax \text{ 与 } x^T Bx \text{ 有相同的正、负惯性指数};$$

③、 A 与 B 相似 $\Leftrightarrow P^{-1}AP = B$;

4, 相似一定合同、合同未必相似;

$$\text{若 } C \text{ 为正交矩阵, 则 } C^T AC = B \Rightarrow A \sim B$$

5, A 为对称阵, 则 A 为二次型矩阵;

6, n 元二次型 $x^T Ax$ 为正定:

$$\Leftrightarrow A \text{ 的正惯性指数为 } n;$$

$$\Leftrightarrow A \text{ 与 } E \text{ 合同, 即存在可逆矩阵 } C, \text{ 使 } C^T AC = E;$$

$$\Leftrightarrow A \text{ 的所有特征值均为正数};$$

$$\Leftrightarrow A \text{ 的各阶顺序主子式均大于 } 0;$$

$$\Rightarrow a_{ii} > 0, |A| > 0; \text{ (必要条件)}$$

第二部分 例题精选

第一章

1. 利用对角线法则计算下列三阶行列式:

$$(1) \begin{vmatrix} 2 & 0 & 1 \\ 1 & -4 & -1 \\ -1 & 8 & 3 \end{vmatrix}; (2) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}; (3) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}; (4) \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}.$$

$$\text{解}(1) = 2 \times (-4) \times 3 + 0 \times (-1) \times (-1) + 1 \times 1 \times 8 - 2 \times (-1) \times 8 - 0 \times 1 \times 3 - 1 \times (-4) \times (-1) = -4$$

$$(2) = acb + bac + cba - a^3 - b^3 - c^3 = 3abc - a^3 - b^3 - c^3$$

$$(3) = bc^2 + ca^2 + ab^2 - cb^2 - ac^2 - ba^2 = (b-a)(c-a)(c-b)$$

$$(4) = x(x+y)y + yx(x+y) + (x+y)yx - x^3 - y^3 - (x+y)^3 = -2(x^3 + y^3)$$

2. 按自然数从小到大为标准次序, 求下列各排列的逆序数:

$$(1) 1 \ 2 \ 3 \ 4; (2) 4 \ 1 \ 3 \ 2; (3) 3 \ 4 \ 2 \ 1; (4) 2 \ 4 \ 1 \ 3;$$

$$(5) 1 \ 3 \dots (2n-1) \ 2 \ 4 \dots (2n); (6) 1 \ 3 \dots (2n-1) \ (2n) \ (2n-2) \dots 2$$

$$\text{解}(1) \text{逆序数} t = 0; (2) \text{逆序数} t = 0 + 1 + 1 + 2 = 4$$

$$(3) \text{逆序数} t = 0 + 0 + 2 + 3 = 5; (4) \text{逆序数} t = 0 + 0 + 2 + 1 = 3$$

$$(5) \text{逆序数} t = 0 + \dots + 0 + (n-1) + (n-2) + \dots + 1 = \frac{n(n-1)}{2}$$

$$(6) \text{逆序数} t = 0 + \dots + 0 + 2 + 4 + \dots + (2n-2) = n(n-1)$$

3. 写出四阶行列式中含有因子 $a_{11}a_{23}$ 的项.

$$\text{解} -a_{11}a_{23}a_{32}a_{44} \text{ 和 } a_{11}a_{23}a_{34}a_{42}$$

4. 计算下列各行列式:

$$(1) \begin{vmatrix} 4 & 1 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 10 & 5 & 2 & 0 \\ 0 & 1 & 1 & 7 \end{vmatrix}; (2) \begin{vmatrix} 2 & 1 & 4 & 1 \\ 3 & -1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 5 & 0 & 6 & 2 \end{vmatrix}; (3) \begin{vmatrix} -ab & ac & ae \\ bd & -cd & de \\ bf & cf & -ef \end{vmatrix}; (4) \begin{vmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix}$$

$$\text{解}(1) \begin{vmatrix} 4 & 1 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 10 & 5 & 2 & 0 \\ 0 & 1 & 1 & 7 \end{vmatrix} \xrightarrow{c_2 - c_3} \begin{vmatrix} 4 & -3 & 2 & -4 \\ 1 & 0 & 0 & 0 \\ 10 & 5 & 2 & -20 \\ 0 & -6 & 1 & 7 \end{vmatrix} = (-1)^{2+1} \begin{vmatrix} -3 & 2 & -4 \\ 5 & 2 & -20 \\ -6 & 1 & 7 \end{vmatrix} \xrightarrow{r_2 - r_1} \begin{vmatrix} -3 & 2 & -4 \\ 8 & 0 & -16 \\ -6 & 1 & 7 \end{vmatrix} \xrightarrow{r_1 - 2r_2} \begin{vmatrix} 9 & 0 & -18 \\ 8 & 0 & -16 \\ -6 & 1 & 7 \end{vmatrix} = 0$$

$$(2) \begin{vmatrix} 2 & 1 & 4 & 1 \\ 3 & -1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 5 & 0 & 6 & 2 \end{vmatrix} \xrightarrow{c_4 - c_2} \begin{vmatrix} 2 & 1 & 4 & 0 \\ 3 & -1 & 2 & 2 \\ 1 & 2 & 3 & 0 \\ 5 & 0 & 6 & 2 \end{vmatrix} \xrightarrow{r_4 - r_2} \begin{vmatrix} 2 & 1 & 4 & 0 \\ 3 & -1 & 2 & 2 \\ 1 & 2 & 3 & 0 \\ 2 & 1 & 4 & 0 \end{vmatrix} = 0$$

$$(3) \begin{vmatrix} -ab & ac & ae \\ bd & -cd & de \\ bf & cf & -ef \end{vmatrix} = adf \begin{vmatrix} b & c & e \\ b & -c & e \\ b & c & -e \end{vmatrix} = adfbce \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 4abcdef$$

$$(4) \begin{vmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix} \xrightarrow{r_1 + ar_2} \begin{vmatrix} 0 & 1+ab & a & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix} = (-1)(-1)^{2+1} \begin{vmatrix} 1+ab & a & 0 \\ -1 & c & 1 \\ 0 & -1 & d \end{vmatrix}$$

$$\xrightarrow{c_3 + dc_2} \begin{vmatrix} 1+ab & a & ad \\ -1 & c & 1+cd \\ 0 & -1 & 0 \end{vmatrix} = (-1)(-1)^{3+2} \begin{vmatrix} 1+ab & ad \\ -1 & 1+cd \end{vmatrix} = abcd + ab + cd + ad + 1$$

5. 求解下列方程:

$$(1) \begin{vmatrix} x+1 & 2 & -1 \\ 2 & x+1 & 1 \\ -1 & 1 & x+1 \end{vmatrix} = 0; \quad (2) \begin{vmatrix} 1 & 1 & 1 & 1 \\ x & a & b & c \\ x^2 & a^2 & b^2 & c^2 \\ x^3 & a^3 & b^3 & c^3 \end{vmatrix} = 0.$$

$$\text{解: (1)} \begin{vmatrix} x+1 & 2 & -1 \\ 2 & x+1 & 1 \\ -1 & 1 & x+1 \end{vmatrix} \xrightarrow{c_3 \leftrightarrow c_1} \begin{vmatrix} x+1 & x+3 & -1 \\ 2 & x+3 & 1 \\ -1 & 0 & x+1 \end{vmatrix} = (x+3) \begin{vmatrix} x+1 & 1 & -1 \\ 2 & 1 & 1 \\ -1 & 0 & x+1 \end{vmatrix}$$

$$\xrightarrow{c_3 \leftrightarrow c_1} = (x+3) \begin{vmatrix} x+1 & 1 & -1 \\ 1-x & 0 & 2 \\ -1 & 0 & x+1 \end{vmatrix} = -(x+3) \begin{vmatrix} 1-x & 2 \\ -1 & x+1 \end{vmatrix} = -(x+3)(3-x^2) = 0, \therefore x = -3, \pm\sqrt{3}$$

$$(2) \begin{vmatrix} 1 & 1 & 1 & 1 \\ x & a & b & c \\ x^2 & a^2 & b^2 & c^2 \\ x^3 & a^3 & b^3 & c^3 \end{vmatrix} = (a-x)(b-x)(c-x)(b-a)(c-a)(c-b) = 0$$

若 a, b, c 三个数中有至少有两个相等, 则 x 可取任何值;

若 a, b, c 三个数两两不等, 则 $x = a$ 或 b 或 c

6. 证明:

$$(1) \begin{vmatrix} a^2 & ab & b^2 \\ 2a & a+b & 2b \\ 1 & 1 & 1 \end{vmatrix} = (a-b)^3; \quad (2) \begin{vmatrix} ax+by & ay+bz & az+bx \\ ay+bz & az+bx & ax+by \\ az+bx & ax+by & ay+bz \end{vmatrix} = (a^3+b^3) \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix};$$

$$(3) \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix} = 0;$$

$$(4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^4 & b^4 & c^4 & d^4 \end{vmatrix} = (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)(a+b+c+d)$$

$$(5) \begin{vmatrix} x & -1 & 0 & \cdots & 0 & 0 \\ 0 & x & -1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & x & -1 \\ a_0 & a_1 & a_2 & \cdots & a_{n-1} & a_n \end{vmatrix} = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

$$\text{证明(1)} \begin{vmatrix} a^2 & ab & b^2 \\ 2a & a+b & 2b \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{c_2 \leftrightarrow c_1} \begin{vmatrix} a^2 & ab-a^2 & b^2-a^2 \\ 2a & b-a & 2b-2a \\ 1 & 0 & 0 \end{vmatrix} \xrightarrow{c_3 \leftrightarrow c_1} = (-1)^{2+1} \begin{vmatrix} ab-a^2 & b^2-a^2 \\ b-a & 2b-2a \end{vmatrix} \\ = (b-a)(b-a) \begin{vmatrix} a & b+a \\ 1 & 2 \end{vmatrix} = (a-b)^3$$

$$(2) \begin{vmatrix} ax+by & ay+bz & az+bx \\ ay+bz & az+bx & ax+by \\ az+bx & ax+by & ay+bz \end{vmatrix} = \begin{vmatrix} ax & ay & az \\ ay & az & ax \\ az & ax & ay \end{vmatrix} + \begin{vmatrix} by & bz & bx \\ bz & bx & by \\ bx & by & bz \end{vmatrix} = (a^3+b^3) \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix}$$

$$(3) \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix} \xrightarrow[\substack{c_3-c_2 \\ c_4-c_3}]{\substack{c_2-c_1 \\ c_3-c_2}} \begin{vmatrix} a^2 & 2a+1 & 2a+3 & 2a+5 \\ b^2 & 2b+1 & 2b+3 & 2b+5 \\ c^2 & 2c+1 & 2c+3 & 2c+5 \\ d^2 & 2d+1 & 2d+3 & 2d+5 \end{vmatrix} \xrightarrow[\substack{c_3-c_2 \\ c_4-c_3}]{c_4-c_3} \begin{vmatrix} a^2 & 2a+1 & 2 & 2 \\ b^2 & 2b+1 & 2 & 2 \\ c^2 & 2c+1 & 2 & 2 \\ d^2 & 2d+1 & 2 & 2 \end{vmatrix} = 0$$

$$(4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^4 & b^4 & c^4 & d^4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ a & b-a & c-a & d-a \\ a^2 & b^2-a^2 & c^2-a^2 & d^2-a^2 \\ a^4 & b^4-a^4 & c^4-a^4 & d^4-a^4 \end{vmatrix}$$

$$= \begin{vmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^4-a^4 & c^4-a^4 & d^4-a^4 \end{vmatrix} = \begin{vmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^2(b^2-a^2) & c^2(c^2-a^2) & d^2(d^2-a^2) \end{vmatrix}$$

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b+a & c+a & d+a \\ b^2(b+a) & c^2(c+a) & d^2(d+a) \end{vmatrix}$$

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 0 & 0 \\ b+a & c-b & d-b \\ b^2(b+a) & c^2(c+a)-b^2(b+a) & d^2(d+a)-b^2(b+a) \end{vmatrix}$$

$$= (b-a)(c-a)(d-a)(c-b)(d-b) \begin{vmatrix} 1 & 1 \\ (c^2+bc+b^2)+a(c+b) & (d^2+bd+b^2)+a(d+b) \end{vmatrix}$$

$$= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)(a+b+c+d)$$

$$(5) D_n = \begin{vmatrix} x & -1 & 0 & \cdots & 0 & 0 \\ 0 & x & -1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & x & -1 \\ a_0 & a_1 & a_2 & \cdots & a_{n-1} & a_n \end{vmatrix} \xrightarrow{\text{按第一列展开}} xD_{n-1} + a_0(-1)^{n+1}(-1)^{n-1}$$

$$= xD_{n-1} + a_0 = x(xD_{n-2} + a_1) + a_0 = x^2D_{n-2} + a_1x + a_0 = \cdots = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

7. 设 n 阶行列式 $D = \det(a_{ij})$, 把 D 上下翻转、或逆时针旋转 90° 、或依副对角线翻转

$$\text{依次得 } D_1 = \begin{vmatrix} a_{n1} & \cdots & a_{nn} \\ \vdots & & \vdots \\ a_{11} & \cdots & a_{1n} \end{vmatrix}, D_2 = \begin{vmatrix} a_{1n} & \cdots & a_{nn} \\ \vdots & & \vdots \\ a_{11} & \cdots & a_{n1} \end{vmatrix}, D_3 = \begin{vmatrix} a_{nn} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{11} \end{vmatrix},$$

$$\text{证明 } D_1 = D_2 = (-1)^{\frac{n(n-1)}{2}} D, D_3 = D.$$

证(1) 将 D 的最后一行经过 $n-1$ 次相邻对换换到第一行, 得到 D_{11} ;

将 D_{11} 的最后一行经过 $n-2$ 次相邻对换换到第二行, 得到 D_{12} ;

.....

$$\text{总共经 } (n-1) + (n-2) + \cdots + 1 = \frac{n(n-1)}{2} \text{ 次相邻对换得到 } D, \therefore D_1 = (-1)^{\frac{n(n-1)}{2}} D$$

(2) 将 D 的最后一行经过 $n-1$ 次相邻对换换到第一行, 得到 D_{21} ;

将 D_{21} 的最后一行经过 $n-2$ 次相邻对换换到第二行, 得到 D_{22} ;

.....

$$\text{总共经 } (n-1) + (n-2) + \cdots + 1 = \frac{n(n-1)}{2} \text{ 次相邻对换得到 } D^T,$$

$$\therefore D_2 = (-1)^{\frac{n(n-1)}{2}} D^T = (-1)^{\frac{n(n-1)}{2}} D$$

8. 计算下列各行列式(D_k 为 k 阶行列式):

$$(1) D_n = \begin{vmatrix} a & & 1 \\ & \ddots & \\ 1 & & a \end{vmatrix}, \text{其中对角线上元素都是} a, \text{未写出的元素都是} 0; \quad (2) D_n = \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix};$$

$$(3) D_{n+1} = \begin{vmatrix} a^n & (a-1)^n & \cdots & (a-n)^n \\ a^{n-1} & (a-1)^{n-1} & \cdots & (a-n)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a & a-1 & \cdots & a-n \\ 1 & 1 & \cdots & 1 \end{vmatrix}; \quad (4) D_{2n} = \begin{vmatrix} a_n & & & b_n \\ & \ddots & & \\ & & a_1 & b_1 \\ & & c_1 & d_1 \\ & & & \ddots \\ c_n & & & & d_n \end{vmatrix},$$

其中未写出的元素都是0;

$$(5) D_n = \det(a_{ij}), \text{其中} a_{ij} = |i-j|; \quad (6) D_n = \begin{vmatrix} 1+a_1 & 1 & \cdots & 1 \\ 1 & 1+a_2 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1+a_n \end{vmatrix}, \text{其中} a_1 a_2 \cdots a_n \neq 0.$$

$$\text{解}(1) D_n = \begin{vmatrix} a & & 1 \\ & \ddots & \\ 1 & & a \end{vmatrix} \xrightarrow{r_1+r_n(1+a)} \begin{vmatrix} 1 & & 1 \\ & \ddots & \\ 1 & & a \end{vmatrix} \xrightarrow{r_n-r_1(1+a)} \begin{vmatrix} 1 & & 1 \\ & \ddots & \\ 0 & & a-1 \end{vmatrix} = (1+a)(a-1)a^{n-2} = a^n - a^{n-2}$$

$$(2) D_n = \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix} \xrightarrow{r_1+r_2+\cdots+r_n[(n-1)a+x]} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix}$$

$$\xrightarrow{i=2, \dots, n} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ 0 & x-a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & x-a \end{vmatrix} = [(n-1)a+x](x-a)^{n-1}$$

$$(3) D_{n+1} = \begin{vmatrix} a^n & (a-1)^n & \cdots & (a-n)^n \\ a^{n-1} & (a-1)^{n-1} & \cdots & (a-n)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a & a-1 & \cdots & a-n \\ 1 & 1 & \cdots & 1 \end{vmatrix} = (-1)^{n+(n-1)+\cdots+1} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ a & (a-1) & \cdots & (a-n) \\ \vdots & \vdots & \ddots & \vdots \\ a^{n-1} & (a-1)^{n-1} & \cdots & (a-n)^{n-1} \\ a^n & (a-1)^n & \cdots & (a-n)^n \end{vmatrix}$$

$$= (-1)^{\frac{n(n+1)}{2}} \prod_{n+1 \geq i > j \geq 1} [a-i+1 - (a-j+1)] = (-1)^{\frac{n(n+1)}{2}} \prod_{n+1 \geq i > j \geq 1} (j-i) = \prod_{n+1 \geq i > j \geq 1} (i-j)$$

$$(4) D_{2n} = \begin{vmatrix} a_n & & & b_n \\ & \ddots & & \\ & & a_1 & b_1 \\ & & c_1 & d_1 \\ & & & \ddots \\ c_n & & & & d_n \end{vmatrix} = a_n \begin{vmatrix} a_{n-1} & & & b_{n-1} \\ & \ddots & & \\ & & a_1 & b_1 \\ & & c_1 & d_1 \\ & & & \ddots \\ c_{n-1} & & & & d_{n-1} \\ 0 & & & & 0 & d_n \end{vmatrix}$$

$$+ (-1)^{2n+1} b_n \begin{vmatrix} 0 & a_{n-1} & & & b_{n-1} & 0 \\ & \ddots & & & \\ & & a_1 & b_1 \\ & & c_1 & d_1 \\ & & & \ddots \\ c_{n-1} & & & & d_{n-1} \\ c_n & & & & 0 & 0 \end{vmatrix} = a_n d_n D_{2n-2} - b_n c_n D_{2n-2} = (a_n d_n - b_n c_n) D_{2n-2}$$

$$\text{于是} D_{2n} = (a_n d_n - b_n c_n) D_{2n-2} = \cdots = \prod_{i=1}^n (a_i d_i - b_i c_i)$$

$$(5) D_n = \det(a_{ij}) = \begin{vmatrix} 0 & 1 & 2 & 3 & \cdots & n-1 \\ 1 & 0 & 1 & 2 & \cdots & n-2 \\ 2 & 1 & 0 & 1 & \cdots & n-3 \\ 3 & 2 & 1 & 0 & \cdots & n-4 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ n-1 & n-2 & n-3 & n-4 & \cdots & 0 \end{vmatrix} \xrightarrow{r_i - r_{i+1}} \begin{vmatrix} -1 & 1 & 1 & 1 & \cdots & 1 \\ -1 & -1 & 1 & 1 & \cdots & 1 \\ -1 & -1 & -1 & 1 & \cdots & 1 \\ -1 & -1 & -1 & -1 & \cdots & 1 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ n-1 & n-2 & n-3 & n-4 & \cdots & 0 \end{vmatrix}$$

$$\xrightarrow{c_i + c_1} \begin{vmatrix} -1 & 0 & 0 & 0 & \cdots & 0 \\ -1 & -2 & 0 & 0 & \cdots & 0 \\ -1 & -2 & -2 & 0 & \cdots & 0 \\ -1 & -2 & -2 & -2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ n-1 & 2n-3 & 2n-4 & 2n-5 & \cdots & n-1 \end{vmatrix} = (-1)^{n-1} (n-1) 2^{n-2}$$

$$(6) D_n = \begin{vmatrix} 1+a_1 & 1 & \cdots & 1 \\ 1 & 1+a_2 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1+a_n \end{vmatrix} = \begin{vmatrix} 1+a_1 & 1+0 & \cdots & 1+0 \\ 1+0 & 1+a_2 & \cdots & 1+0 \\ \vdots & \vdots & \ddots & \vdots \\ 1+0 & 1+0 & \cdots & 1+a_n \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \cdots & 0 \\ 1 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & a_n \end{vmatrix} + \begin{vmatrix} a_1 & 1 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \cdots & a_n \end{vmatrix} + \cdots + \begin{vmatrix} a_1 & 0 & \cdots & 1 \\ 0 & a_2 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{vmatrix} + \begin{vmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{vmatrix}$$

$$= a_1 a_2 \cdots a_n (1 + a_1 a_2 \cdots a_n + a_1 a_2 \cdots a_n + \cdots + a_1 a_2 \cdots a_n + a_1 a_2 \cdots a_n) = \left(\prod_{i=1}^n a_i \right) \left(1 + \sum_{i=1}^n 1/a_i \right)$$

9. 设 $D = \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 1 & 3 & -4 \\ 2 & 0 & 1 & -1 \\ 1 & -5 & 3 & -3 \end{vmatrix}$, D 的 (i, j) 元的代数余子式记作 A_{ij} , 求 $A_{31} + 3A_{32} - 2A_{33} + 2A_{34}$.

$$\text{解: } A_{31} + 3A_{32} - 2A_{33} + 2A_{34} = \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 1 & 3 & -4 \\ 1 & 3 & -2 & 2 \\ 1 & -5 & 3 & -3 \end{vmatrix} = \begin{vmatrix} -2 & 2 & 2 & -2 \\ -5 & 1 & 3 & -4 \\ 1 & 3 & -2 & 2 \\ 1 & -5 & 3 & -3 \end{vmatrix} = \begin{vmatrix} -2 & 0 & 0 & 0 \\ -5 & -4 & -2 & 1 \\ 1 & 4 & -1 & 1 \\ 1 & -4 & 4 & -4 \end{vmatrix}$$

$$= -2 \begin{vmatrix} -4 & -2 & 1 \\ 4 & -1 & 1 \\ -4 & 4 & -4 \end{vmatrix} = -2 \begin{vmatrix} -4 & -2 & 1 \\ 0 & -3 & 2 \\ 0 & 6 & -5 \end{vmatrix} = -2 \begin{vmatrix} -4 & -2 & 1 \\ 0 & -3 & 2 \\ 0 & 0 & -1 \end{vmatrix} = 24$$

10. 用克拉默法则解下列方程组:

$$(1) \begin{cases} x_1 + x_2 + x_3 + x_4 = 5, \\ x_1 + 2x_2 - x_3 + 4x_4 = -2, \\ 2x_1 - 3x_2 - x_3 - 5x_4 = -2, \\ 3x_1 + x_2 + 2x_3 + 11x_4 = 0. \end{cases} \quad (2) \begin{cases} 5x_1 + 6x_2 = 1, \\ x_1 + 5x_2 + 6x_3 = 0, \\ x_2 + 5x_3 + 6x_4 = 0, \\ x_3 + 5x_4 + 6x_5 = 0, \\ x_4 + 5x_5 = 1. \end{cases}$$

$$\text{解(1)} D = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 4 \\ 2 & -3 & -1 & -5 \\ 3 & 1 & 2 & 11 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & -5 & -3 & -7 \\ 0 & -2 & -1 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & -13 & 8 \\ 0 & 0 & -5 & 14 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & -13 & 8 \\ 0 & 0 & -5 & 14 \end{vmatrix} = \begin{vmatrix} -13 & 8 \\ -5 & 14 \end{vmatrix} = -142$$

$$D_1 = \begin{vmatrix} 5 & 1 & 1 & 1 \\ -2 & 2 & -1 & 4 \\ -2 & -3 & -1 & -5 \\ 0 & 1 & 2 & 11 \end{vmatrix} = -142, D_2 = \begin{vmatrix} 1 & 5 & 1 & 1 \\ 1 & -2 & -1 & 4 \\ 2 & -2 & -1 & -5 \\ 3 & 0 & 2 & 11 \end{vmatrix} = -284$$

$$D_3 = \begin{vmatrix} 1 & 1 & 5 & 1 \\ 1 & 2 & -2 & 4 \\ 2 & -3 & -2 & -5 \\ 3 & 1 & 0 & 11 \end{vmatrix} = -426, D_4 = \begin{vmatrix} 1 & 1 & 1 & 5 \\ 1 & 2 & -1 & -2 \\ 2 & -3 & -1 & -2 \\ 3 & 1 & 2 & 0 \end{vmatrix} = 142$$

$$\therefore x_1 = \frac{D_1}{D} = 1, x_2 = \frac{D_2}{D} = 2, x_3 = \frac{D_3}{D} = 3, x_4 = \frac{D_4}{D} = -1$$

$$(2) D = \begin{vmatrix} 5 & 6 & 0 & 0 & 0 \\ 1 & 5 & 6 & 0 & 0 \\ 0 & 1 & 5 & 6 & 0 \\ 0 & 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 1 & 5 \end{vmatrix} = 5 \begin{vmatrix} 5 & 6 & 0 & 0 \\ 1 & 5 & 6 & 0 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 1 & 5 \end{vmatrix} - \begin{vmatrix} 5 & 6 & 0 & 0 \\ 1 & 5 & 6 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 1 & 6 \end{vmatrix}$$

$$= 25 \begin{vmatrix} 5 & 6 & 0 \\ 1 & 5 & 6 \\ 0 & 1 & 5 \end{vmatrix} - 5 \begin{vmatrix} 5 & 6 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 6 \end{vmatrix} - 6 \begin{vmatrix} 5 & 6 & 0 \\ 1 & 5 & 6 \\ 0 & 1 & 5 \end{vmatrix} + 19 \begin{vmatrix} 5 & 6 & 0 \\ 1 & 5 & 6 \\ 0 & 1 & 5 \end{vmatrix} - 30 \begin{vmatrix} 5 & 6 \\ 1 & 5 \end{vmatrix}$$

$$= 95 \begin{vmatrix} 5 & 6 \\ 1 & 5 \end{vmatrix} - 19 \begin{vmatrix} 5 & 0 \\ 1 & 6 \end{vmatrix} - 30 \begin{vmatrix} 5 & 6 \\ 1 & 5 \end{vmatrix} = 95 \times 19 - 19 \times 30 - 30 \times 19 = 665$$

$$D_1 = \begin{vmatrix} 1 & 6 & 0 & 0 & 0 \\ 0 & 5 & 6 & 0 & 0 \\ 0 & 1 & 5 & 6 & 0 \\ 0 & 0 & 1 & 5 & 6 \\ 1 & 0 & 0 & 1 & 5 \end{vmatrix} = 1507, D_2 = \begin{vmatrix} 5 & 1 & 0 & 0 & 0 \\ 1 & 0 & 6 & 0 & 0 \\ 0 & 0 & 5 & 6 & 0 \\ 0 & 0 & 1 & 5 & 6 \\ 0 & 1 & 0 & 1 & 5 \end{vmatrix} = -1145$$

$$D_3 = \begin{vmatrix} 5 & 6 & 1 & 0 & 0 \\ 1 & 5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 6 & 0 \\ 0 & 0 & 0 & 5 & 6 \\ 0 & 0 & 1 & 1 & 5 \end{vmatrix} = 703, D_4 = \begin{vmatrix} 5 & 6 & 0 & 1 & 0 \\ 1 & 5 & 6 & 0 & 0 \\ 0 & 1 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & 5 \end{vmatrix} = -395, D_5 = \begin{vmatrix} 5 & 6 & 0 & 0 & 1 \\ 1 & 5 & 6 & 0 & 0 \\ 0 & 1 & 5 & 6 & 0 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix} = 212$$

$$\therefore x_1 = \frac{1507}{665}, x_2 = -\frac{1145}{665}, x_3 = \frac{703}{665}, x_4 = \frac{-395}{665}, x_5 = \frac{212}{665}$$

11. 问 λ 取何值时, 齐次线性方程组

$$\begin{cases} \lambda x_1 + x_2 + x_3 = 0 \\ x_1 + \mu x_2 + x_3 = 0 \\ x_1 + 2\mu x_2 + x_3 = 0 \end{cases} \text{ 有非零解?}$$

$$\text{解 当系数行列式 } D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \mu & 1 \\ 1 & 2\mu & 1 \end{vmatrix} = \mu(1-\lambda) = 0,$$

即 $\mu = 0$ 或 $\lambda = 1$ 时齐次线性方程组有非零解.

12. 问 λ 取何值时, 齐次线性方程组

$$\begin{cases} (1-\lambda)x_1 - 2x_2 + 4x_3 = 0 \\ 2x_1 + (3-\lambda)x_2 + x_3 = 0 \\ x_1 + x_2 + (1-\lambda)x_3 = 0 \end{cases} \text{ 有非零解?}$$

$$\text{解 当系数行列式 } D = \begin{vmatrix} 1-\lambda & -2 & 4 \\ 2 & 3-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = \lambda(\lambda-3)(2-\lambda) = 0,$$

即 $\lambda = 0$ 或 $\lambda = 3$ 或 $\lambda = 2$ 时齐次线性方程组有非零解.

1. 计算下列乘积:

$$(1) \begin{pmatrix} 4 & 3 & 1 \\ 1 & -2 & 3 \\ 5 & 7 & 0 \end{pmatrix} \begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix}; \quad (2) (1 \ 2 \ 3) \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}; \quad (3) \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} -1 & 2 \end{pmatrix};$$

$$(4) \begin{pmatrix} 2 & 1 & 4 & 0 \\ 1 & -1 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & -1 & 2 \\ 1 & -3 & 1 \\ 4 & 0 & -2 \end{pmatrix}; \quad (5) (x_1, x_2, x_3) \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

$$\text{解: } (1) \begin{pmatrix} 4 & 3 & 1 \\ 1 & -2 & 3 \\ 5 & 7 & 0 \end{pmatrix} \begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 7 + 3 \times 2 + 1 \times 1 \\ 1 \times 7 + (-2) \times 2 + 3 \times 1 \\ 5 \times 7 + 7 \times 2 + 0 \times 1 \end{pmatrix} = \begin{pmatrix} 35 \\ 6 \\ 49 \end{pmatrix}$$

$$(2) (1 \ 2 \ 3) \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = (1 \times 3 + 2 \times 2 + 3 \times 1) = (10)$$

$$(3) \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 \times (-1) & 2 \times 2 \\ 1 \times (-1) & 1 \times 2 \\ 3 \times (-1) & 3 \times 2 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ -1 & 2 \\ -3 & 6 \end{pmatrix}$$

$$(4) \begin{pmatrix} 2 & 1 & 4 & 0 \\ 1 & -1 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & -1 & 2 \\ 1 & -3 & 1 \\ 4 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 6 & -7 & 8 \\ 20 & -5 & -6 \end{pmatrix}$$

$$(5) (x_1, x_2, x_3) \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= (a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \quad a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \quad a_{31}x_1 + a_{32}x_2 + a_{33}x_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + 2a_{12}x_1x_2 + 2a_{13}x_1x_3 + 2a_{23}x_2x_3$$

$$2. \text{ 设 } A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix} \text{ 求 } 3AB - 2A \text{ 及 } A^T B.$$

$$\text{解: } 3AB - 2A = 3 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix} - 2 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

$$= 3 \begin{pmatrix} 0 & 5 & 8 \\ 0 & -5 & 6 \\ 2 & 9 & 0 \end{pmatrix} - \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & -2 \\ 2 & -2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 15 & 24 \\ 0 & -15 & 18 \\ 6 & 27 & 0 \end{pmatrix} - \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & -2 \\ 2 & -2 & 2 \end{pmatrix} = \begin{pmatrix} -2 & 13 & 22 \\ -2 & -17 & 20 \\ 4 & 29 & -2 \end{pmatrix}$$

$$A^T B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & 4 \\ 0 & 5 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 5 & 8 \\ 0 & -5 & 6 \\ 2 & 9 & 0 \end{pmatrix}$$

$$3. \text{ 已知两个线性变换 } \begin{cases} x_1 = 2y_1 + y_3 \\ x_2 = -2y_1 + 3y_2 + 2y_3 \\ x_3 = 4y_1 + y_2 + 5y_3 \end{cases}, \quad \begin{cases} y_1 = -3z_1 + z_2 \\ y_2 = 2z_1 + z_3 \\ y_3 = -z_2 + 3z_3 \end{cases},$$

求从 z_1, z_2, z_3 到 x_1, x_2, x_3 的线性变换.

$$\text{解: } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ -2 & 3 & 2 \\ 4 & 1 & 5 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ -2 & 3 & 2 \\ 4 & 1 & 5 \end{pmatrix} \begin{pmatrix} -3 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & -1 & 3 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} -6 & 1 & 3 \\ 12 & -4 & 9 \\ -10 & -1 & 16 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

$$\therefore \begin{cases} x_1 = -6z_1 + z_2 + 3z_3 \\ x_2 = 12z_1 - 4z_2 + 9z_3 \\ x_3 = -10z_1 - z_2 + 16z_3 \end{cases}$$

$$4. \text{ 设 } A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \text{ 问:}$$

(1) $AB = BA$ 吗? (2) $(A+B)^2 = A^2 + 2AB + B^2$ 吗? (3) $(A+B)(A-B) = A^2 - B^2$ 吗?

$$\text{解: (1)} AB = \begin{pmatrix} 3 & 4 \\ 4 & 6 \end{pmatrix} \quad BA = \begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix} \quad \therefore AB \neq BA$$

$$(2) (A+B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2 (\because AB \neq BA)$$

$$(3) (A+B)(A-B) = A^2 - AB + BA - B^2 \neq A^2 - B^2 (\because AB \neq BA)$$

5. 举反例说明下列命题是错误的:

(1) 若 $A^2 = 0$, 则 $A = 0$; (2) 若 $A^2 = A$, 则 $A = 0$ 或 $A = E$; (3) $AX = AY$, 且 $A \neq 0$, 则 $X = Y$.

$$\text{解: (1)} A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad A^2 = 0, \text{ 但 } A \neq 0; \quad (2) A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad A^2 = A, \text{ 但 } A \neq 0 \text{ 且 } A \neq E;$$

$$(3) A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad X = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad Y = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad AX = AY, \text{ 且 } A \neq 0, \text{ 但 } X \neq Y.$$

$$6. \text{ 设 } A = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}, \text{ 求 } A^2, A^3, \dots, A^n.$$

$$\text{解: } A^2 = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2\lambda & 1 \end{pmatrix} \quad A^3 = A^2 A = \begin{pmatrix} 1 & 0 \\ 2\lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 3\lambda & 1 \end{pmatrix}$$

利用数学归纳法证明: $A^k = \begin{pmatrix} 1 & 0 \\ k\lambda & 1 \end{pmatrix}$ 当 $k=1$ 时, 显然成立; 假设 k 时成立, 则当 $k+1$ 时

$$A^{k+1} = A^k A = \begin{pmatrix} 1 & 0 \\ k\lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ (k+1)\lambda & 1 \end{pmatrix} \quad (\text{证毕})$$

$$7. \text{ 设 } A = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}, \text{ 求 } A^k.$$

$$\text{解: } A^2 = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{pmatrix}$$

$$A^3 = A^2 A = \begin{pmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda^3 & 3\lambda^2 & 3\lambda \\ 0 & \lambda^3 & 3\lambda^2 \\ 0 & 0 & \lambda^3 \end{pmatrix}$$

$$\text{利用数学归纳法证明: } A^k = \begin{pmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{pmatrix}$$

当 $k=2$ 时, 显然成立. 假设 k 成立, 则 $k+1$ 时,

$$A^{k+1} = A^k A = \begin{pmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} \lambda^{k+1} & (k+1)\lambda^k & \frac{k(k+1)}{2}\lambda^{k-1} \\ 0 & \lambda^{k+1} & (k+1)\lambda^k \\ 0 & 0 & \lambda^{k+1} \end{pmatrix} \quad (\text{证毕})$$

8. 设 A, B 为 n 阶矩阵, 且 A 为对称阵, 证明 $B^T A B$ 也是对称阵.

证明 $(B^T A B)^T = B^T (B^T A)^T = B^T A^T B = B^T A B, \therefore B^T A B$ 也是对称阵.

9. 设 A, B 为 n 阶对称阵, 证明 AB 是对称阵的充分必要条件是 $AB = BA$.

证明 AB 是对称阵 $\Leftrightarrow (AB)^T = AB \Leftrightarrow B^T A^T = AB \Leftrightarrow AB = BA$.

10. 求下列矩阵的逆阵:

$$(1) \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}; \quad (2) \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix};$$

$$(3) \begin{pmatrix} 1 & 2 & -1 \\ 3 & 4 & -2 \\ 5 & -4 & 1 \end{pmatrix}; \quad (4) \begin{pmatrix} a_1 & & 0 \\ & a_2 & \\ & & \ddots \\ 0 & & & a_n \end{pmatrix}, (a_1 a_2 \cdots a_n \neq 0)$$

解: (1) $|A| = 1, A_{11} = 5, A_{12} = -2, A_{21} = -2, A_{22} = 1,$

$$A^* = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}, \quad A^{-1} = \frac{1}{|A|} A^* = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}$$

(2) $|A| = 1, A_{11} = \cos \theta, A_{12} = -\sin \theta, A_{21} = \sin \theta, A_{22} = \cos \theta,$

$$A^* = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \quad A^{-1} = \frac{1}{|A|} A^* = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

(3) $|A| = 2,$

$$A_{11} = -4 \quad A_{12} = -13 \quad A_{13} = -32$$

$$A_{21} = 2 \quad A_{22} = 6 \quad A_{23} = 14$$

$$A_{31} = 0 \quad A_{32} = -1 \quad A_{33} = -2,$$

$$A^{-1} = \frac{1}{|A|} A^* = \begin{pmatrix} -2 & 1 & 0 \\ -13 & 3 & -1 \\ 2 & 7 & -1 \end{pmatrix}$$

$$(4) \begin{pmatrix} a_1 & & 0 \\ & a_2 & \\ & & \ddots \\ 0 & & & a_n \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{a_1} & & 0 \\ & \frac{1}{a_2} & \\ & & \ddots \\ 0 & & & \frac{1}{a_n} \end{pmatrix}$$

11. 解下列矩阵方程:

$$(1) \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} X = \begin{pmatrix} 4 & -6 \\ 2 & 1 \end{pmatrix}; \quad (2) X \begin{pmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{pmatrix};$$

$$(3) \begin{pmatrix} 1 & 4 \\ -1 & 2 \end{pmatrix} X \begin{pmatrix} 2 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 0 & -1 \end{pmatrix}; \quad (4) \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} X \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{pmatrix}$$

$$\text{解: (1)} X = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}^{-1} \begin{pmatrix} 4 & -6 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 4 & -6 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -23 \\ 0 & 8 \end{pmatrix}$$

$$(2) X = \begin{pmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 & 3 \\ 4 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ -2 & 3 & -2 \\ -3 & 3 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 2 & 1 \\ -8 & 5 & -2 \\ 3 & 3 & 0 \end{pmatrix}$$

$$(3) X = \begin{pmatrix} 1 & 4 \\ -1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ -1 & 1 \end{pmatrix}^{-1} = \frac{1}{12} \begin{pmatrix} 2 & -4 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & 0 \end{pmatrix}$$

$$(4) X = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -4 & 3 \\ 2 & 0 & -1 \\ 1 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -4 \\ 1 & 0 & -2 \end{pmatrix}$$

12. 利用逆阵解下列线性方程组:

$$(1) \begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 2x_1 + 2x_2 + 5x_3 = 2 \\ 3x_1 + 5x_2 + x_3 = 3 \end{cases} \quad (2) \begin{cases} x_1 - x_2 - x_3 = 2 \\ 2x_1 - x_2 - 3x_3 = 1 \\ 3x_1 + 2x_2 - 5x_3 = 0 \end{cases}$$

$$\text{解: } (1) \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 5 \\ 3 & 5 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 5 \\ 3 & 5 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x_1 = 1 \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

$$(2) \begin{pmatrix} 1 & -1 & -1 \\ 2 & -1 & -3 \\ 3 & 2 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & -1 & -1 \\ 2 & -1 & -3 \\ 3 & 2 & -5 \end{pmatrix}^{-1} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix} \Rightarrow \begin{cases} x_1 = 5 \\ x_2 = 0 \\ x_3 = 3 \end{cases}$$

13. 已知线性变换 $\begin{cases} x_1 = 2y_1 + 2y_2 + y_3 \\ x_2 = 3y_1 + y_2 + 5y_3 \\ x_3 = 3y_1 + 2y_2 + 3y_3 \end{cases}$, 求从变量 x_1, x_2, x_3 到变量 y_1, y_2, y_3 的线性变换.

$$\text{解: } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \Rightarrow \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{pmatrix}^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -7 & -4 & 9 \\ 6 & 3 & -7 \\ 3 & 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\therefore \begin{cases} y_1 = -7x_1 - 4x_2 + 9x_3 \\ y_2 = 6x_1 + 3x_2 - 7x_3 \\ y_3 = 3x_1 + 2x_2 - 4x_3 \end{cases}$$

14. 设 A 为 3 阶矩阵, $|A| = \frac{1}{2}$, 求 $|(2A)^{-1} - 5A^*|$.

$$\begin{aligned} \text{证明 } |(2A)^{-1} - 5A^*| &= \left| \frac{1}{2}A^{-1} - 5A^* \right| = \left| \frac{1}{2}A^{-1} - 5|A|A^{-1} \right| \\ &= \left| \frac{1}{2}A^{-1} - \frac{5}{2}A^{-1} \right| = |-2A^{-1}| = (-2)^3 |A^{-1}| = -8 \times 2 = -16. \end{aligned}$$

15. 设 $A = \begin{pmatrix} 0 & 3 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{pmatrix}$, $AB = A + 2B$, 求 B .

$$\begin{aligned} \text{解: } AB &= A + 2B \Rightarrow (A - 2E)B = A \\ \Rightarrow B &= (A - 2E)^{-1}A = \begin{pmatrix} -2 & 3 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 3 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} -1 & 3 & 3 \\ -1 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 3 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 3 \\ -1 & 2 & 3 \\ 1 & 1 & 0 \end{pmatrix} \end{aligned}$$

16. 设 $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, 且 $AB + E = A^2 + B$, 求 B .

$$\begin{aligned} \text{解: } AB + E &= A^2 + B \\ \Rightarrow (A - E)B &= A^2 - E = (A - E)(A + E), \\ \text{而 } A - E &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \text{ 可逆, } \therefore B = A + E = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{pmatrix} \end{aligned}$$

17. 设 $A = \text{diag}(1, -2, 1)$, $A^*BA = 2BA - 8E$, 求 B .

$$\begin{aligned} \text{解: } A^*BA &= 2BA - 8E \Rightarrow A(A^*BA)A^{-1} = A(2BA - 8E)A^{-1} \Rightarrow |A|B = 2AB - 8E \Rightarrow (2A - |A|E)B = 8E, \\ \text{由 } |A| &= -2, \quad 2A - |A|E = 2(A + E) = \text{diag}(4, -2, 4) \text{ 可逆} \\ \Rightarrow B &= 8\text{diag}(4, -2, 4)^{-1} = 8\text{diag}\left(\frac{1}{4}, -\frac{1}{2}, \frac{1}{4}\right) = \text{diag}(2, -4, 2) = 2A \end{aligned}$$

18. 已知矩阵 A 的伴随阵 $A^* = \text{diag}(1, 1, 1, 8)$, 且 $ABA^{-1} = BA^{-1} + 3E$, 求 B .

解: $ABA^{-1} = BA^{-1} + 3E \Rightarrow A^*(ABA^{-1})A = A^*(BA^{-1} + 3E)A \Rightarrow |A|B = A^*B + 3|A|E$,

而 $|A^*| = 8 = |A|^{4-1} \Rightarrow |A| = 2 \Rightarrow (2E - A^*)B = 6E \Rightarrow B = 6(2E - A^*)^{-1}$

$$= 6 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 3 & 0 & -6 \end{pmatrix}^{-1} = \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 6 & 0 & 6 & 0 \\ 0 & 3 & 0 & -1 \end{pmatrix}$$

19. 设 $P^{-1}AP = \Lambda$, 其中 $P = \begin{pmatrix} -1 & -4 \\ 1 & 1 \end{pmatrix}$, $\Lambda = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$, 求 A^{11} .

解: $P^{-1}AP = \Lambda \Rightarrow A = P\Lambda P^{-1}$

$$\Rightarrow A^{11} = P\Lambda^{11}P^{-1} = \begin{pmatrix} -1 & -4 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} (-1)^{11} & 0 \\ 0 & 2^{11} \end{pmatrix} \begin{pmatrix} \frac{1}{3} & \frac{4}{3} \\ -\frac{1}{3} & -\frac{1}{3} \end{pmatrix}$$

$$= \begin{pmatrix} 2731 & 2732 \\ -683 & -684 \end{pmatrix}$$

20. 设 $AP = P\Lambda$, 其中 $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}$, $\Lambda = \begin{pmatrix} -1 & & \\ & 1 & \\ & & 5 \end{pmatrix}$, 求 $\varphi(A) = A^6(5E - 6A + A^2)$.

解: $AP = P\Lambda \Rightarrow A = P\Lambda P^{-1}$

$$\Rightarrow \varphi(A) = P\varphi(\Lambda)P^{-1} = P \begin{pmatrix} \varphi(-1) & & \\ & \varphi(1) & \\ & & \varphi(5) \end{pmatrix} P^{-1}$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 5 - 6 \times (-1) + 1 & & \\ & 5 - 6 \times 1 + 1 & \\ & & 5^6(5 - 6 \times 5 + 5^2) \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}^{-1}$$

$$= \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 12 & & \\ & 0 & \\ & & 0 \end{pmatrix} \begin{pmatrix} 2 & 2 & 2 \\ 3 & 0 & -3 \\ 1 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix}$$

21. 设 $A^k = 0$ (k 为正整数), 证明

$$(E - A)^{-1} = E + A + A^2 + \cdots + A^{k-1}.$$

(证一) $\because A^k = 0$

$$\therefore (E - A)(E + A + A^2 + \cdots + A^{k-1}) = E - A + A - A^2 + \cdots - A^{k-1} + A^{k-1} - A^k = E$$

$$\therefore (E - A)^{-1} = E + A + A^2 + \cdots + A^{k-1}$$

(证二) $\because A^k = 0$

$$\therefore E = (E - A) + (A - A^2) + (A^2 - A^3) + \cdots + (A^{k-1} - A^k)$$

$$= (E - A)(E + A + A^2 + \cdots + A^{k-1})$$

$$\therefore (E - A)^{-1} = E + A + A^2 + \cdots + A^{k-1}$$

22. 设方阵 A 满足 $A^2 - A - 2E = 0$, 证明 A 及 $A + 2E$ 都可逆, 并求 A^{-1} 及 $(A + 2E)^{-1}$.

$$\text{证明 } A^2 - A - 2E = 0 \Rightarrow A^2 - A = 2E \Rightarrow A[\frac{1}{2}(A - E)] = E \Rightarrow A \text{ 可逆}, A^{-1} = \frac{1}{2}(A - E)$$

$$A^2 - A - 2E = 0 \Rightarrow A + 2E = A^2, \because A \text{ 可逆} \therefore A^2 \text{ 可逆},$$

$$\text{且 } (A^2)^{-1} = (A^{-1})^2 = [\frac{1}{2}(A - E)]^2 = \frac{1}{4}(3E - A), \therefore (A + 2E)^{-1} = (A^2)^{-1} = \frac{1}{4}(3E - A)$$

23. 设矩阵 A 可逆, 证明其伴随矩阵 A^* 也可逆, 且 $(A^*)^{-1} = (A^{-1})^*$.

$$\text{证明 } \because A^*(A^{-1})^* = A^*(A^{-1})^* A^{-1} A = A^* |A^{-1}| EA = |A^{-1}| A^* A = |A^{-1}| |A| E = E$$

$$\therefore A^* \text{ 也可逆, 且 } (A^*)^{-1} = (A^{-1})^*.$$

24. 设 n 阶矩阵 A 的伴随阵为 A^* , 证明: (1) 若 $|A| = 0$, 则 $|A^*| = 0$; (2) $|A^*| = |A|^{n-1}$.

证明(1)用反证法: 假设 $|A^*| \neq 0$, 则有 $A^*(A^*)^{-1} = E$, 又 $A = AA^*(A^*)^{-1} = |A|(A^*)^{-1} = 0$, 从而 $A^* = 0$, 这与 $|A^*| \neq 0$ 矛盾, 故假设不成立, $|A^*| = 0$;

(2) 若 $|A| = 0$, 则 $|A^*| = 0, \therefore |A^*| = |A|^{n-1}$

若 $|A| \neq 0$, 由 $AA^* = |A|E$ 得, $|A||A^*| = |AA^*| = ||A|E| = |A|^n, \therefore |A^*| = |A|^{n-1}$.

25. 计算
$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & -3 \end{pmatrix}.$$

解:
$$\begin{pmatrix} A_1 & E \\ 0 & A_2 \end{pmatrix} \begin{pmatrix} E & B_1 \\ 0 & B_2 \end{pmatrix} = \begin{pmatrix} A_1 & A_1 B_1 + B_2 \\ 0 & A_2 B_2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 5 & 2 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & -4 & 3 \\ 0 & 0 & 0 & -9 \end{pmatrix}$$

26. 设 $A = \begin{pmatrix} 3 & 4 & 0 \\ 4 & -3 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}$, 求 $|A^6|$ 及 A^4 .

解: $|A| = \begin{vmatrix} 3 & 4 \\ 4 & -3 \end{vmatrix} \begin{vmatrix} 2 & 0 \\ 2 & 2 \end{vmatrix} = -100, |A^6| = |A|^6 = (-100)^6 = 10^{16}$

$$A^4 = \begin{pmatrix} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}^4 & 0 \\ 0 & \begin{pmatrix} 2 & 0 \\ 2 & 2 \end{pmatrix}^4 \end{pmatrix} = \begin{pmatrix} 5^4 & 0 & 0 \\ 0 & 5^4 & 0 \\ 0 & 2^4 & 0 \\ 0 & 2^6 & 2^4 \end{pmatrix}$$

27. 设 n 阶矩阵 A 及 s 阶矩阵 B 都可逆, 求

(1) $\begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}^{-1}$, (2) $\begin{pmatrix} A & 0 \\ C & B \end{pmatrix}^{-1}$.

解: (1) 设 $\begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}^{-1} = \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix}$, 则 $\begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix} \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix} = E = \begin{pmatrix} E & 0 \\ 0 & E \end{pmatrix}$

$$\Rightarrow \begin{cases} AX_3 = E \\ AX_4 = 0 \\ BX_1 = 0 \\ BX_2 = E \end{cases} \Rightarrow \begin{cases} X_3 = A^{-1} \\ X_4 = 0 \\ X_1 = 0 \\ X_2 = B^{-1} \end{cases} \Rightarrow \begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & B^{-1} \\ A^{-1} & 0 \end{pmatrix}$$

(2) 设 $\begin{pmatrix} A & 0 \\ C & B \end{pmatrix}^{-1} = \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix}$, 则 $\begin{pmatrix} A & 0 \\ C & B \end{pmatrix} \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix} = E = \begin{pmatrix} E & 0 \\ 0 & E \end{pmatrix}$

$$\Rightarrow \begin{cases} AX_1 = E \\ AX_2 = 0 \\ CX_1 + BX_3 = 0 \\ CX_2 + BX_4 = E \end{cases} \Rightarrow \begin{cases} X_1 = A^{-1} \\ X_2 = 0 \\ X_3 = -B^{-1}CA^{-1} \\ X_4 = B^{-1} \end{cases} \Rightarrow \begin{pmatrix} A & 0 \\ C & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & 0 \\ -B^{-1}CA^{-1} & B^{-1} \end{pmatrix}.$$

28. 求下列矩阵的逆阵:

(1) $\begin{pmatrix} 5 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 5 & 2 \end{pmatrix}$; (2) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 2 & 1 & 3 & 0 \\ 1 & 2 & 1 & 4 \end{pmatrix}.$

解: (1)
$$\begin{pmatrix} 5 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 5 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}^{-1} & 0 \\ 0 & \begin{pmatrix} 8 & 3 \\ 5 & 2 \end{pmatrix}^{-1} \end{pmatrix} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ -2 & 5 & 0 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & -5 & 8 \end{pmatrix}$$

(2)
$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 2 & 1 & 3 & 0 \\ 1 & 2 & 1 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}^{-1} & 0 \\ -\begin{pmatrix} 3 & 0 \\ 1 & 4 \end{pmatrix}^{-1} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}^{-1} & \begin{pmatrix} 3 & 0 \\ 1 & 4 \end{pmatrix}^{-1} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ -\frac{1}{3} & -\frac{1}{6} & \frac{1}{3} & 0 \\ \frac{1}{8} & -\frac{5}{24} & -\frac{1}{12} & \frac{1}{4} \end{pmatrix}$$

第三章

1. 用初等行变换把下列矩阵化为行最简形矩阵:

$$(1) \begin{pmatrix} 1 & 0 & 2 & -1 \\ 2 & 0 & 3 & 1 \\ 3 & 0 & 4 & 3 \end{pmatrix}; (2) \begin{pmatrix} 0 & 2 & -3 & 1 \\ 0 & 3 & -4 & 3 \\ 0 & 4 & -7 & -1 \end{pmatrix}; (3) \begin{pmatrix} 1 & -1 & 3 & -4 & 3 \\ 3 & -3 & 5 & -4 & 1 \\ 2 & -2 & 3 & -2 & 0 \\ 3 & -3 & 4 & -2 & -1 \end{pmatrix}; (4) \begin{pmatrix} 2 & 3 & 1 & -3 & -7 \\ 1 & 2 & 0 & -2 & -4 \\ 3 & -2 & 8 & 3 & 0 \\ 2 & -3 & 7 & 4 & 3 \end{pmatrix}$$

$$\text{解: } (1) \begin{pmatrix} 1 & 0 & 2 & -1 \\ 2 & 0 & 3 & 1 \\ 3 & 0 & 4 & 3 \end{pmatrix} \xrightarrow{\substack{r_2-2r_1 \\ r_3-3r_1}} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -2 & 6 \end{pmatrix} \xrightarrow{r_3+2r_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(2) \begin{pmatrix} 0 & 2 & -3 & 1 \\ 0 & 3 & -4 & 3 \\ 0 & 4 & -7 & -1 \end{pmatrix} \xrightarrow{\substack{r_1 \leftrightarrow r_2 \\ r_3-r_2}} \begin{pmatrix} 0 & 3 & -4 & 3 \\ 0 & 2 & -3 & 1 \\ 0 & 1 & -1 & 2 \end{pmatrix} \xrightarrow{\substack{r_2 \leftrightarrow r_3 \\ r_1-3r_3}} \begin{pmatrix} 0 & 1 & -1 & 2 \\ 0 & 2 & -3 & 1 \\ 0 & 3 & -4 & 3 \end{pmatrix} \xrightarrow{\substack{r_2-2r_1 \\ r_3-3r_1}} \begin{pmatrix} 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & -3 \end{pmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \begin{pmatrix} 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & -1 & -5 \end{pmatrix} \xrightarrow{r_2+r_3} \begin{pmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & -8 \end{pmatrix} \xrightarrow{r_2 \times (-1/8)} \begin{pmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{r_1+r_2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{r_1+3r_2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$(3) \begin{pmatrix} 1 & -1 & 3 & -4 & 3 \\ 3 & -3 & 5 & -4 & 1 \\ 2 & -2 & 3 & -2 & 0 \\ 3 & -3 & 4 & -2 & -1 \end{pmatrix} \xrightarrow{\substack{r_2-r_1 \\ r_3-2r_1 \\ r_4-3r_1}} \begin{pmatrix} 1 & -1 & 3 & -4 & 3 \\ 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & -3 & 6 & -6 \\ 0 & 0 & -5 & 10 & -10 \end{pmatrix} \xrightarrow{\substack{r_3+3r_2 \\ r_4+5r_2}} \begin{pmatrix} 1 & -1 & 3 & -4 & 3 \\ 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1+r_2} \begin{pmatrix} 1 & 0 & 4 & -6 & 5 \\ 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-4r_2} \begin{pmatrix} 1 & 0 & 0 & 2 & -3 \\ 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(4) \begin{pmatrix} 2 & 3 & 1 & -3 & -7 \\ 1 & 2 & 0 & -2 & -4 \\ 3 & -2 & 8 & 3 & 0 \\ 2 & -3 & 7 & 4 & 3 \end{pmatrix} \xrightarrow{\substack{r_1 \leftrightarrow r_2 \\ r_3-r_1 \\ r_4-r_1}} \begin{pmatrix} 1 & 2 & 0 & -2 & -4 \\ 2 & 3 & 1 & -3 & -7 \\ 0 & -2 & 8 & 5 & 4 \\ 0 & -5 & 7 & 6 & 7 \end{pmatrix} \xrightarrow{\substack{r_2-2r_1 \\ r_3+2r_1 \\ r_4+5r_1}} \begin{pmatrix} 1 & 2 & 0 & -2 & -4 \\ 0 & -1 & 1 & 1 & 1 \\ 0 & 0 & 8 & 9 & 12 \\ 0 & 0 & 7 & 8 & 11 \end{pmatrix} \xrightarrow{\substack{r_2 \times (-1) \\ r_3-r_2 \\ r_4-r_2}} \begin{pmatrix} 1 & 2 & 0 & -2 & -4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 7 & 9 & 12 \\ 0 & 0 & 7 & 8 & 11 \end{pmatrix} \xrightarrow{\substack{r_3 \times (1/7) \\ r_4-r_3}} \begin{pmatrix} 1 & 2 & 0 & -2 & -4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-2r_2} \begin{pmatrix} 1 & 0 & 0 & -2 & -4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1+2r_3} \begin{pmatrix} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

2. 设 $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 \end{pmatrix}$, 求一个可逆阵 P , 使 PA 为行最简形.

$$\text{解: } A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 \end{pmatrix} \xrightarrow{r_2-2r_1} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 5 & 4 & 3 & 2 \end{pmatrix} \xrightarrow{r_3-5r_1} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & -6 & -12 & -18 \end{pmatrix} \xrightarrow{r_3+6r_2} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{r_2 \times (-1)} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1-2r_2} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = B$$

$$P = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -6 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -1 & 0 \\ 7 & -6 & 1 \end{pmatrix}$$

3. 设 $A = \begin{pmatrix} -5 & 3 & 1 \\ 2 & -1 & 1 \end{pmatrix}$, (1) 求可逆阵 P , 使 PA 为行最简形; (2) 求一个可逆矩阵 Q , 使 QA^T 为行最简形.

$$\text{解: } (1) A = \begin{pmatrix} -5 & 3 & 1 \\ 2 & -1 & 1 \end{pmatrix} \xrightarrow{r_1+2r_2} \begin{pmatrix} -1 & 1 & 3 \\ 2 & -1 & 1 \end{pmatrix} \xrightarrow{r_2+2r_1} \begin{pmatrix} -1 & 1 & 3 \\ 0 & 1 & 7 \end{pmatrix} \xrightarrow{r_1 \times (-1)} \begin{pmatrix} 1 & -1 & -3 \\ 0 & 1 & 7 \end{pmatrix} \xrightarrow{r_1+r_2} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 7 \end{pmatrix} \xrightarrow{r_1-4r_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{pmatrix} \xrightarrow{r_2 \times (1/7)} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$P = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$$

$$(2) A^T = \begin{pmatrix} -5 & 2 \\ 3 & -1 \\ 1 & 1 \end{pmatrix} \xrightarrow{r_1+2r_2} \begin{pmatrix} 1 & 0 \\ 3 & -1 \\ 1 & 1 \end{pmatrix} \xrightarrow{r_2-3r_1} \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{pmatrix} \xrightarrow{r_2 \times (-1)} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \xrightarrow{r_3-r_2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \xrightarrow{r_2 \times (-1)} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 5 & 0 \\ -4 & -7 & 1 \end{pmatrix}$$

4. 利用矩阵的初等变换, 求下列方阵的逆阵:

$$(1) \begin{pmatrix} 3 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{pmatrix}; (2) \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & -2 \\ 0 & 1 & 2 & 1 \end{pmatrix}.$$

$$\text{解: (1)} \begin{pmatrix} 3 & 2 & 1 & 1 & 0 & 0 \\ 3 & 1 & 5 & 0 & 1 & 0 \\ 3 & 2 & 3 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{r_2-r_1 \\ r_3-r_1}} \begin{pmatrix} 3 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & 4 & -1 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{r_2 \times \frac{1}{-1} \\ r_3+4r_2 \\ r_1+2r_2}} \begin{pmatrix} 3 & 0 & 0 & \frac{7}{2} & 2 & -\frac{9}{2} \\ 0 & -1 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$\xrightarrow{\substack{r_1 \times \frac{1}{3} \\ r_2 \times (-1)}} \begin{pmatrix} 1 & 0 & 0 & \frac{7}{6} & \frac{2}{3} & -\frac{3}{2} \\ 0 & 1 & 0 & -1 & -1 & 2 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}, \therefore \begin{pmatrix} 3 & 2 & 1 \\ 3 & 1 & 5 \\ 3 & 2 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{7}{6} & \frac{2}{3} & -\frac{3}{2} \\ -1 & -1 & 2 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$(2) \begin{pmatrix} 3 & -2 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & -2 & -3 & -2 & 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{r_2 \leftrightarrow r_1 \\ r_3-3r_1}} \begin{pmatrix} 1 & -2 & -3 & -2 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & 9 & 5 & 1 & 0 & -3 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{\substack{r_2 \leftrightarrow r_4 \\ r_3-4r_4 \\ r_1+2r_4 \\ r_2+2r_4}} \begin{pmatrix} 1 & -2 & -3 & -2 & 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & -3 & -4 \\ 0 & 0 & 0 & 1 & 2 & 1 & -6 & -10 \end{pmatrix} \xrightarrow{\substack{r_2-r_4 \\ r_3-r_4 \\ r_1+2r_4}} \begin{pmatrix} 1 & -2 & -3 & 0 & 4 & 2 & -11 & -20 \\ 0 & 1 & 2 & 0 & -2 & -1 & 6 & 11 \\ 0 & 0 & 1 & 0 & -1 & -1 & 3 & 6 \\ 0 & 0 & 0 & 1 & 2 & 1 & -6 & -10 \end{pmatrix}$$

$$\xrightarrow{\substack{r_1-2r_2 \\ r_3+2r_2 \\ r_4+2r_2}} \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & -2 & -4 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 3 & 6 \\ 0 & 0 & 0 & 1 & 2 & 1 & -6 & -10 \end{pmatrix}, \therefore \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & -2 \\ 0 & 1 & 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1 & -2 & -4 \\ 0 & 1 & 0 & -1 \\ -1 & -1 & 3 & 6 \\ 2 & 1 & -6 & -10 \end{pmatrix}$$

$$5.(1) \text{ 设 } A = \begin{pmatrix} 4 & 1 & -2 \\ 2 & 2 & 1 \\ 3 & 1 & -1 \end{pmatrix}, B = \begin{pmatrix} 1 & -3 \\ 2 & 2 \\ 3 & -1 \end{pmatrix}, \text{ 求 } X \text{ 使 } AX = B.$$

$$(2) \text{ 设 } A = \begin{pmatrix} 0 & 2 & 1 \\ 2 & -1 & 3 \\ -3 & 3 & -4 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -3 & 1 \end{pmatrix}, \text{ 求 } X \text{ 使 } XA = B.$$

$$\text{解: (1)} (A \ B) = \begin{pmatrix} 4 & 1 & -2 & 1 & -3 \\ 2 & 2 & 1 & 2 & 2 \\ 3 & 1 & -1 & 3 & -1 \end{pmatrix} \xrightarrow{\substack{r_1 \times 3 \\ r_2 \times 6 \\ r_3 \times 4}} \begin{pmatrix} 12 & 3 & -6 & 3 & -9 \\ 12 & 12 & 6 & 12 & 12 \\ 12 & 4 & -4 & 12 & -4 \end{pmatrix}$$

$$\xrightarrow{\substack{r_2-r_1 \\ r_3-r_1 \\ r_2-3r_3 \\ r_3+(-2)r_3}} \begin{pmatrix} 12 & 3 & -6 & 3 & -9 \\ 0 & 1 & 2 & 9 & 5 \\ 0 & 0 & 1 & 12 & 4 \end{pmatrix} \xrightarrow{\substack{r_1+6r_2 \\ r_2-2r_3 \\ r_1-3r_2 \\ r_3+12r_3}} \begin{pmatrix} 1 & 0 & 0 & 10 & 2 \\ 0 & 1 & 0 & -15 & -3 \\ 0 & 0 & 1 & 12 & 4 \end{pmatrix}, \therefore X = A^{-1}B = \begin{pmatrix} 10 & 2 \\ -15 & -3 \\ 12 & 4 \end{pmatrix}$$

$$(2) (A^T \ B^T) = \begin{pmatrix} 0 & 2 & -3 & 1 & 2 \\ 2 & -1 & 3 & 2 & -3 \\ 1 & 3 & -4 & 3 & 1 \end{pmatrix} \xrightarrow{\substack{r_1 \leftrightarrow r_3 \\ r_2+2r_1 \\ r_3-2r_1}} \begin{pmatrix} 1 & 3 & -4 & 3 & 1 \\ 0 & -7 & 11 & -4 & -5 \\ 0 & 2 & -3 & 1 & 2 \end{pmatrix} \xrightarrow{\substack{r_2 \times (-2) \\ r_3 \times 7}} \begin{pmatrix} 1 & 3 & -4 & 3 & 1 \\ 0 & 14 & -22 & 8 & 10 \\ 0 & 14 & -21 & 7 & 14 \end{pmatrix}$$

$$\xrightarrow{\substack{r_2-r_3 \\ r_1+22r_3 \\ r_2+4r_3 \\ r_3+14r_3}} \begin{pmatrix} 1 & 3 & 0 & -1 & 8 \\ 0 & 1 & 0 & -1 & 7 \\ 0 & 0 & 1 & -1 & 4 \end{pmatrix} \xrightarrow{\substack{r_1-3r_2 \\ r_1+4r_2 \\ r_3+14r_3}} \begin{pmatrix} 1 & 0 & 0 & 2 & -4 \\ 0 & 1 & 0 & -1 & 7 \\ 0 & 0 & 1 & -1 & 4 \end{pmatrix}, \therefore X = BA^{-1} = \begin{pmatrix} 2 & -1 & -1 \\ -4 & 7 & 4 \end{pmatrix}$$

$$6. \text{ 设 } A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix}, AX = 2X + A, \text{ 求 } X.$$

$$\text{解: } AX = 2X + A \Rightarrow (A - 2E)X = A \Rightarrow \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ -1 & 0 & -1 \end{pmatrix} X = A$$

$$\begin{pmatrix} -1 & -1 & 0 & 1 & -1 & 0 \\ 0 & -1 & -1 & 0 & 1 & -1 \\ -1 & 0 & -1 & -1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{pmatrix}, \therefore X = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

7. 在秩是 r 的矩阵中, 有没有等于 0 的 $r-1$ 阶子式? 有没有等于 0 的 r 阶子式? 答: 都可能.

8. 从矩阵 A 中划去一行得到矩阵 B , 问 A, B 的秩的关系怎样? 答: $R(A) \geq R(B)$

9. 求作一个秩是 4 的方阵, 它的两个行向量是 $(1, 0, 1, 0, 0), (1, -1, 0, 0, 0)$.

$$\text{答: } \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

10. 求下列矩阵的秩, 并求一个最高阶非零子式:

$$(1) \begin{pmatrix} 3 & 1 & 0 & 2 \\ 1 & -1 & 2 & -1 \\ 1 & 3 & -4 & 4 \end{pmatrix}; (2) \begin{pmatrix} 3 & 2 & -1 & -3 & -1 \\ 2 & -1 & 3 & 1 & -3 \\ 7 & 0 & 5 & -1 & -8 \end{pmatrix}; (3) \begin{pmatrix} 2 & 1 & 8 & 3 & 7 \\ 2 & -3 & 0 & 7 & -5 \\ 3 & -2 & 5 & 8 & 0 \\ 1 & 0 & 3 & 2 & 0 \end{pmatrix}$$

$$\text{解: } (1) A = \begin{pmatrix} 3 & 1 & 0 & 2 \\ 1 & -1 & 2 & -1 \\ 1 & 3 & -4 & 4 \end{pmatrix} \xrightarrow{\substack{r_1 \leftrightarrow r_2 \\ r_2 - r_1 \\ r_3 - r_1}} \begin{pmatrix} 1 & -1 & 2 & -1 \\ 0 & 4 & -6 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \therefore R(A) = 2,$$

$$\therefore \begin{vmatrix} 3 & 1 \\ 1 & -1 \end{vmatrix} = -4 \neq 0, \therefore \begin{vmatrix} 3 & 1 \\ 1 & -1 \end{vmatrix} \text{ 是 } A \text{ 的一个最高阶非零子式}$$

$$(2) A = \begin{pmatrix} 3 & 2 & -1 & -3 & -1 \\ 2 & -1 & 3 & 1 & -3 \\ 7 & 0 & 5 & -1 & -8 \end{pmatrix} \xrightarrow{\substack{r_1 \leftrightarrow r_2 \\ r_2 \leftrightarrow r_1 \\ r_3 - 2r_1 \\ r_3 - 3r_2 \\ r_3 + 4r_2}} \begin{pmatrix} 2 & -1 & 3 & 1 & -3 \\ 3 & 2 & -1 & -3 & -1 \\ 0 & 7 & -11 & -9 & 7 \\ 0 & 0 & 0 & 0 & -6 \end{pmatrix},$$

$$\therefore R(A) = 3, \therefore \begin{vmatrix} 3 & 2 & -1 \\ 2 & -1 & -3 \\ 7 & 0 & -8 \end{vmatrix} \text{ 是 } A \text{ 的一个最高阶非零子式}$$

$$(3) A = \begin{pmatrix} 2 & 1 & 8 & 3 & 7 \\ 2 & -3 & 0 & 7 & -5 \\ 3 & -2 & 5 & 8 & 0 \\ 1 & 0 & 3 & 2 & 0 \end{pmatrix} \xrightarrow{\substack{r_1 \leftrightarrow r_4 \\ r_2 - r_1 \\ r_3 - 2r_1 \\ r_4 - 3r_1 \\ r_4 + 2r_2}} \begin{pmatrix} 1 & 0 & 3 & 2 & 0 \\ 0 & -3 & -6 & 3 & -5 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 4 & 8 & -4 & 12 \end{pmatrix} \xrightarrow{\substack{r_2 \leftrightarrow r_3 \\ r_4 + 3r_2 \\ r_4 - 4r_3}} \begin{pmatrix} 1 & 0 & 3 & 2 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 & 12 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 3 & 2 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\therefore R(A) = 3, \therefore \begin{vmatrix} 2 & 1 & 7 \\ 2 & -3 & -5 \\ 1 & 0 & 0 \end{vmatrix} = 16 \neq 0, \therefore \begin{vmatrix} 2 & 1 & 7 \\ 2 & -3 & -5 \\ 1 & 0 & 0 \end{vmatrix} \text{ 是 } A \text{ 的一个最高阶非零子式}$$

11. 设 A, B 都是 $m \times n$ 矩阵, 证明 $A \sim B$ 的充分必要条件是 $R(A) = R(B)$.

证明 " $\Rightarrow$ " 见本章定理3

" $\Leftarrow$ " A, B 都是 $m \times n$ 矩阵, 且 $R(A) = R(B) = r$, 则 A, B 具有相同的标准型 $\begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}_{m \times n}$

即 $A \sim \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}_{m \times n}, B \sim \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}_{m \times n}$, 所以 $A \sim B$.

12. 设 $A = \begin{pmatrix} 1 & -2 & 3k \\ -1 & 2k & -3 \\ k & -2 & 3 \end{pmatrix}$, 问 k 为何值, 可使 (1) $R(A) = 1$; (2) $R(A) = 2$; (3) $R(A) = 3$.

$$\text{解 } A = \begin{pmatrix} 1 & -2 & 3k \\ -1 & 2k & -3 \\ k & -2 & 3 \end{pmatrix} \xrightarrow{\substack{r_2 + r_1 \\ r_3 - r_1}} \begin{pmatrix} 1 & -2 & 3k \\ 0 & 2(k-1) & 3(k-1) \\ 0 & 0 & 3(1-k)(2+k) \end{pmatrix} \quad \begin{array}{l} (1) k=1 \text{ 时, } R(A)=1; \\ (2) k=-2 \text{ 时, } R(A)=2; \\ (3) k \neq 1 \text{ 且 } k \neq -2 \text{ 时, } R(A)=3. \end{array}$$

13. 求解下列齐次线性方程组:

$$(1) \begin{cases} x_1 + x_2 + 2x_3 - x_4 = 0 \\ 2x_1 + x_2 + x_3 - x_4 = 0 \\ 2x_1 + 2x_2 + x_3 + 2x_4 = 0 \end{cases} \quad (2) \begin{cases} x_1 + 2x_2 + x_3 - x_4 = 0 \\ 3x_1 + 6x_2 - x_3 - 3x_4 = 0 \\ 5x_1 + 10x_2 + x_3 - 5x_4 = 0 \end{cases}$$

$$(3) \begin{cases} 2x_1 + 3x_2 - x_3 + 5x_4 = 0 \\ 3x_1 + x_2 + 2x_3 - 7x_4 = 0 \\ 4x_1 + x_2 - 3x_3 + 6x_4 = 0 \\ x_1 - 2x_2 + 4x_3 - 7x_4 = 0 \end{cases} \quad (3) \begin{cases} 3x_1 + 4x_2 - 5x_3 + 7x_4 = 0 \\ 2x_1 - 3x_2 + 3x_3 - 2x_4 = 0 \\ 4x_1 + 11x_2 - 13x_3 + 16x_4 = 0 \\ 7x_1 - 2x_2 + x_3 + 3x_4 = 0 \end{cases}$$

解: (1)对系数矩阵施行行初等变换,有

$$\begin{pmatrix} 1 & 1 & 2 & -1 \\ 2 & 1 & 1 & -1 \\ 2 & 2 & 1 & 2 \end{pmatrix} \xrightarrow[r_2-r_1]{r_3-r_1} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -3 \\ 0 & 0 & -3 & 4 \end{pmatrix} \xrightarrow[r_2 \times (-1)]{r_2 \times (-\frac{1}{3})} \begin{pmatrix} 1 & 0 & 0 & -\frac{4}{3} \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -\frac{4}{3} \end{pmatrix} \therefore \begin{cases} x_1 = \frac{4}{3}x_4 \\ x_2 = -3x_4 \\ x_3 = \frac{4}{3}x_4 \end{cases}, x_4 \text{ 为自由变量}$$

$$\text{故通解为 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = c \begin{pmatrix} \frac{4}{3} \\ -3 \\ \frac{4}{3} \\ 1 \end{pmatrix}, (c \in R)$$

(2)对系数矩阵施行行初等变换,有

$$\begin{pmatrix} 1 & 2 & 1 & -1 \\ 3 & 6 & -1 & -3 \\ 5 & 10 & 1 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \therefore \begin{cases} x_1 = -2x_2 + x_4 \\ x_3 = 0 \end{cases}, x_2, x_4 \text{ 为自由变量}$$

$$\text{故通解为 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = c_1 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, (c_1, c_2 \in R)$$

(3)对系数矩阵施行行初等变换,有

$$\begin{pmatrix} 2 & 3 & -1 & 5 \\ 3 & 1 & 2 & -7 \\ 4 & 1 & -3 & 6 \\ 1 & -2 & 4 & -7 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \therefore \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases} \text{ 只有零解}$$

(4)对系数矩阵施行行初等变换,有

$$\begin{pmatrix} 3 & 4 & -5 & 7 \\ 2 & -3 & 3 & -2 \\ 4 & 11 & -13 & 16 \\ 7 & -2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -\frac{3}{17} & \frac{3}{17} \\ 0 & 1 & -\frac{9}{17} & \frac{20}{17} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \therefore \begin{cases} x_1 = \frac{3}{17}x_3 - \frac{3}{17}x_4 \\ x_2 = \frac{9}{17}x_3 - \frac{20}{17}x_4 \end{cases}, x_3, x_4 \text{ 为自由变量}$$

$$\text{故通解为 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = c_1 \begin{pmatrix} \frac{3}{17} \\ \frac{9}{17} \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -\frac{3}{17} \\ -\frac{20}{17} \\ 0 \\ 1 \end{pmatrix}, (c_1, c_2 \in R)$$

14. 求解下列非齐次线性方程组

$$(1) \begin{cases} 4x_1 + 2x_2 - x_3 = 2, \\ 3x_1 - x_2 + 2x_3 = 10, \\ 11x_1 + 3x_2 = 8; \end{cases} (2) \begin{cases} 2x + 3y + z = 4, \\ x - 2y + 4z = -5, \\ 3x + 8y - 2z = 13, \\ 4x - y + 9z = -6; \end{cases} (3) \begin{cases} 2x + y - z + w = 1, \\ 4x + 2y - 2z + w = 2, \\ 2x + y - z - w = 1; \end{cases} (4) \begin{cases} 2x + y - z + w = 1, \\ 3x - 2y + z - 3w = 4, \\ x + 4y - 3z + 5w = -2. \end{cases}$$

解: (1) 对增广矩阵B施行初等行变换

$$B = \begin{pmatrix} 4 & 2 & -1 & 2 \\ 3 & -1 & 2 & 10 \\ 11 & 3 & 0 & 8 \end{pmatrix} \xrightarrow[r_3-r_1]{r_1 \leftrightarrow r_3} \begin{pmatrix} 11 & 3 & 0 & 8 \\ 4 & 2 & -1 & 2 \\ 3 & -1 & 2 & 10 \end{pmatrix} \xrightarrow[r_2-3r_1]{r_3-3r_1} \begin{pmatrix} 11 & 3 & 0 & 8 \\ 0 & -7 & -3 & -22 \\ 0 & -10 & 2 & -2 \end{pmatrix} \xrightarrow[r_2 \times (-1)]{r_3 \times (-1)} \begin{pmatrix} 11 & 3 & 0 & 8 \\ 0 & 7 & 3 & 22 \\ 0 & 10 & -2 & 2 \end{pmatrix}$$

$R(A) = 2, R(B) = 3, R(A) \neq R(B)$, 故方程组无解.

$$(2) B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & -2 & 4 & -5 \\ 3 & 8 & -2 & 13 \\ 4 & -1 & 9 & -6 \end{pmatrix} \xrightarrow[r_4-r_1]{r_2 \leftrightarrow r_1} \begin{pmatrix} 1 & -2 & 4 & -5 \\ 2 & 3 & 1 & 4 \\ 3 & 8 & -2 & 13 \\ 4 & -1 & 9 & -6 \end{pmatrix} \xrightarrow[r_4-4r_1]{r_3-3r_1} \begin{pmatrix} 1 & -2 & 4 & -5 \\ 0 & 7 & -7 & 14 \\ 0 & 14 & -14 & 28 \\ 0 & 9 & -25 & 26 \end{pmatrix} \xrightarrow[r_4-r_2]{r_3 \times (-1)} \begin{pmatrix} 1 & -2 & 4 & -5 \\ 0 & 7 & -7 & 14 \\ 0 & -14 & 14 & -28 \\ 0 & 9 & -25 & 26 \end{pmatrix} \xrightarrow[r_4 \times (-1)]{r_2 \times (-1)} \begin{pmatrix} 1 & -2 & 4 & -5 \\ 0 & -7 & 7 & -14 \\ 0 & -14 & 14 & -28 \\ 0 & -9 & 25 & -26 \end{pmatrix}$$

$$\text{的线性方程组为 } \begin{cases} x + 2z = -1 \\ y - z = 2 \\ z = z \end{cases}, \text{ 通解为: } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = c \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}, (c \in R)$$

$$(3) B = \begin{pmatrix} 2 & 1 & -1 & 1 & 1 \\ 4 & 2 & -2 & 1 & 2 \\ 2 & 1 & -1 & -1 & 1 \end{pmatrix} \xrightarrow[r_3-r_1]{r_2-r_1} \begin{pmatrix} 2 & 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[r_2 \times (-1)]{r_1 \times (-1)} \begin{pmatrix} -1 & -1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[r_1+r_2]{r_1 \times (-1)} \begin{pmatrix} -1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[r_1 \times (-1)]{r_1 \times (-1)} \begin{pmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \therefore \begin{cases} x = -\frac{1}{2}y + \frac{1}{2}z - \frac{1}{2} \\ w = 0 \end{cases}, y, z \text{ 为自由变量}$$

$$\therefore \text{通解为} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = c_1 \begin{pmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 0 \end{pmatrix}, (c_1, c_2 \in \mathbb{R})$$

$$(4) B = \begin{pmatrix} 2 & 1 & -1 & 1 & 1 \\ 3 & -2 & 1 & -3 & 4 \\ 1 & 4 & -3 & 5 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -\frac{1}{7} & -\frac{1}{7} & \frac{6}{7} \\ 0 & 1 & -\frac{5}{7} & \frac{9}{7} & -\frac{5}{7} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, R(A) = R(B) = 2 < 4, \text{故} \begin{cases} x = \frac{1}{7}z + \frac{1}{7}w + \frac{6}{7} \\ y = \frac{5}{7}z - \frac{9}{7}w - \frac{5}{7} \\ z, w \text{ 为自由变量} \end{cases}$$

$$\therefore \text{通解为} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = c_1 \begin{pmatrix} \frac{1}{7} \\ \frac{5}{7} \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} \frac{1}{7} \\ -\frac{9}{7} \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{6}{7} \\ -\frac{5}{7} \\ 0 \\ 0 \end{pmatrix}, (c_1, c_2 \in \mathbb{R})$$

15. 写出一个以 $x = c_1 \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 4 \\ 0 \\ 1 \end{pmatrix}$ 为通解的齐次线性方程组.

$$\text{解} \begin{cases} x_2 + 3x_3 - 4x_4 = 0 \\ x_1 - 2x_3 + 2x_4 = 0 \end{cases}$$

16. λ 取何值时, 非齐次线性方程组 $\begin{cases} \lambda x_1 + x_2 + x_3 = 1 \\ x_1 + \lambda x_2 + x_3 = \lambda \\ x_1 + x_2 + \lambda x_3 = \lambda^2 \end{cases}$ (1) 有唯一解; (2) 无解; (3) 有无穷多解;

解 因系数矩阵 A 是方阵, 方程组有唯一解的充要条件是系数行列式 $|A| = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda+2)(\lambda-1)^2 \neq 0$,

因此当 $\lambda \neq -2$ 且 $\lambda \neq 1$ 时方程组有唯一解;

当 $\lambda = -2$ 时, 增广矩阵 $B = \begin{pmatrix} -2 & 1 & 1 & 1 \\ 1 & -2 & 1 & -2 \\ 1 & 1 & -2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 1 & -2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$\therefore R(A) = 2 < R(B) = 3$, 方程组无解;

当 $\lambda = 1$ 时, 增广矩阵 $B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $\therefore R(A) = R(B) = 1 < 3$ 方程组有无穷多解.

17. 非齐次线性方程组 $\begin{cases} -2x_1 + x_2 + x_3 = -2 \\ x_1 - 2x_2 + x_3 = \lambda \\ x_1 + x_2 - 2x_3 = \lambda^2 \end{cases}$, 当 λ 取何值时有解? 并求出它的通解.

解 $B = \begin{pmatrix} -2 & 1 & 1 & -2 \\ 1 & -2 & 1 & \lambda \\ 1 & 1 & -2 & \lambda^2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 1 & \lambda \\ 0 & -3 & 3 & 2(\lambda-1) \\ 0 & 0 & 0 & (\lambda-1)(\lambda+2) \end{pmatrix}$

方程组有解, 只需 $(\lambda-1)(\lambda+2) = 0$, 得 $\lambda = 1$ 或 $\lambda = -2$.

当 $\lambda = 1$ 时 $B \sim \begin{pmatrix} 1 & -2 & 1 & 1 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, 通解为 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, (c \in \mathbb{R})$

当 $\lambda = -2$ 时 $B \sim \begin{pmatrix} 1 & -2 & 1 & -2 \\ 0 & -3 & 3 & -6 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, 通解为 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}, (c \in \mathbb{R})$

18. 设 $\begin{cases} (2-\lambda)x_1 + 2x_2 - 2x_3 = 1, \\ 2x_1 + (5-\lambda)x_2 - 4x_3 = 2, \\ -2x_1 - 4x_2 + (5-\lambda)x_3 = -\lambda-1, \end{cases}$ 问 λ 为何值时,

此方程组有唯一解、无解或有无穷多解?并在有无穷多解时求其通解.

解 当 $|A| = \begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & 5-\lambda & -4 \\ -2 & -4 & 5-\lambda \end{vmatrix} = -(\lambda-1)^2(\lambda-10) \neq 0$, 即 $\lambda \neq 1$ 且 $\lambda \neq 10$ 时方程组有唯一解;

当 $\lambda=10$ 时 $B = \begin{pmatrix} -8 & 2 & -2 & 1 \\ 2 & -5 & -4 & 2 \\ -2 & -4 & -5 & -11 \end{pmatrix} \sim \begin{pmatrix} 2 & -5 & -4 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, $R(A) < R(B)$, 方程组无解;

当 $\lambda=1$ 时 $B = \begin{pmatrix} 1 & 2 & -2 & 1 \\ 2 & 4 & -4 & 2 \\ -2 & -4 & 4 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $R(A) = R(B) < 3$, 方程组有无穷多解.

通解 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = c_1 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, ($c_1, c_2 \in R$).

19. 证明 $R(A)=1$ 的充分必要条件是存在非零列向量 a 及非零行向量 b^T , 使 $A=ab^T$.

证明 " $\Rightarrow$ " 若 $R(A)=1$, 则 A 的各行对应成比例, 即

$$A = \begin{pmatrix} k_1 a_{11} & k_1 a_{12} & \cdots & k_1 a_{1n} \\ k_2 a_{11} & k_2 a_{12} & \cdots & k_2 a_{1n} \\ \vdots & \vdots & & \vdots \\ k_n a_{11} & k_n a_{12} & \cdots & k_n a_{1n} \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{pmatrix} \begin{pmatrix} k_1 & k_2 & \cdots & k_n \end{pmatrix},$$

记 $a = (a_{11} \ a_{12} \ \cdots \ a_{1n})^T$, $b = (k_1 \ k_2 \ \cdots \ k_n)^T$, 有 $A=ab^T$, 且 $a \neq 0, b^T \neq 0$,

否则 $A=0, R(A)=0$ 与已知矛盾

" $\Leftarrow$ " 若 $A=ab^T$, 则 $R(A) \leq R(a)=1$, 又可断言 $R(A) > 0$ (否则 $A=0, a, b$ 中至少有一个为零向量, 与已知矛盾.)

于是 $R(A)=1$.

20. 设 A 为列满秩矩阵, $AB=C$, 证明线性方程 $Bx=\theta$ 与 $Cx=\theta$ 同解.

证明: 若 $Bx=\theta$, 则 $ABx=\theta$, 即 $Cx=\theta$;

若 $Cx=\theta$, 即 $ABx=\theta$, A 为列满秩矩阵, 故 $Bx=\theta$,

因此线性方程 $Bx=\theta$ 与 $Cx=\theta$ 同解.

21. 设 A 为 $m \times n$ 矩阵, 证明

方程 $AX=E_m$ 有解的充分必要条件是 $R(A)=m$.

证明

$AX=E_m$ 有解 $\Leftrightarrow R(A)=R(A, E_m) \geq m, R(A) \leq m \Leftrightarrow R(A)=m$;

第四章

1. 已知向量组

$$A: a_1 = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}, a_2 = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 2 \end{pmatrix}, a_3 = \begin{pmatrix} 2 \\ 3 \\ 0 \\ 1 \end{pmatrix}; \quad B: b_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \\ 2 \end{pmatrix}, b_2 = \begin{pmatrix} 0 \\ -2 \\ 1 \\ 1 \end{pmatrix}, b_3 = \begin{pmatrix} 4 \\ 4 \\ 1 \\ 3 \end{pmatrix}.$$

证明 B 组能由 A 组线性表示, 但 A 组不能由 B 组线性表示.

$$\text{证明 } \because (A, B) = \begin{pmatrix} 0 & 3 & 2 & 2 & 0 & 4 \\ 1 & 0 & 3 & 1 & -2 & 4 \\ 2 & 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 1 & 2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 3 & 1 & -2 & 4 \\ 0 & 1 & -6 & -1 & 5 & -7 \\ 0 & 0 & 4 & 1 & -3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\therefore R(A) = R(A, B) = 3, \therefore B$ 组能由 A 组线性表示;

$$\because B = \begin{pmatrix} 2 & 0 & 4 \\ 1 & -2 & 4 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 4 \\ -1 & 5 & -7 \\ 1 & -3 & 5 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\therefore R(B) = 2 < R(A, B) = 3, \therefore A$ 组不能由 B 组线性表示.

2. 已知向量组A: $\alpha_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$; B: $\beta_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \beta_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \beta_3 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$, 证明A组与B组等价.

$$\text{证 } (B \ A) = \begin{pmatrix} -1 & 1 & 3 & 0 & 1 \\ 0 & 2 & 2 & 1 & 1 \\ 1 & 1 & -1 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} -1 & 1 & 3 & 0 & 1 \\ 0 & 2 & 2 & 1 & 1 \\ 0 & 2 & 2 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} -1 & 1 & 3 & 0 & 1 \\ 0 & 2 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \therefore R(B, A) = R(B) = 2$$

又 $R(A) = 2$, $\therefore$ A组与B组等价

3. 已知 $R(a_1, a_2, a_3) = 2, R(a_2, a_3, a_4) = 3$, 证明

(1) a_1 能由 a_2, a_3 线性表示;

(2) a_4 不能由 a_1, a_2, a_3 线性表示.

证明(1) $\because R(a_2, a_3, a_4) = 3, \therefore a_2, a_3, a_4$ 线性无关, $\therefore a_2, a_3$ 线性无关,

又 $\because R(a_1, a_2, a_3) = 2, \therefore a_1, a_2, a_3$ 线性相关, $\therefore a_1$ 能由 a_2, a_3 线性表示;

(2)(反证法) 若 a_4 能由 a_1, a_2, a_3 线性表示, 而 a_1 能由 a_2, a_3 线性表示, 则 a_4 能由 a_2, a_3 线性表示, 则 a_2, a_3, a_4 线性相关, 与已知矛盾. 故 a_4 不能由 a_1, a_2, a_3 线性表示.

4. 判定下列向量组是线性相关还是线性无关:

$$(1) \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}; \quad (2) \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}.$$

$$\text{解: } (1) A = \begin{pmatrix} -1 & 2 & 1 \\ 3 & 1 & 4 \\ 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \therefore R(A) = 2 < 3, \therefore \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} \text{ 线性相关};$$

$$(2) A = \begin{pmatrix} 2 & -1 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \therefore R(A) = 3, \therefore \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \text{ 线性无关}.$$

5. 问 a 取什么值时下列向量组线性相关?

$$\alpha_1 = \begin{pmatrix} a \\ 1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ a \\ -1 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 1 \\ -1 \\ a \end{pmatrix}.$$

$$\text{解一 由 } \begin{vmatrix} a & 1 & 1 \\ 1 & a & -1 \\ 1 & -1 & a \end{vmatrix} = (1+a)^2(a-2) = 0, \text{ 解得}$$

$a = -1$ 或 $a = 2$ 时 $\alpha_1, \alpha_2, \alpha_3$ 线性相关.

$$\text{解二 } \because (\alpha_3, \alpha_2, \alpha_1) = \begin{pmatrix} 1 & 1 & a \\ -1 & a & 1 \\ a & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & a \\ 0 & a+1 & a+1 \\ 0 & -1-a & 1-a^2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & a \\ 0 & a+1 & a+1 \\ 0 & 0 & 2+a-a^2 \end{pmatrix},$$

当 $a+1=0$ 或者 $2+a-a^2=0$ 时 $R(\alpha_3, \alpha_2, \alpha_1) < 3, \therefore a = -1$ 或 $a = 2$ 时 $\alpha_1, \alpha_2, \alpha_3$ 线性相关.

6. 设 α_1, α_2 线性无关, $\alpha_1 + b, \alpha_2 + b$ 线性相关, 求向量 b 用

α_1, α_2 线性表示的表示式.

(解一) $\because \alpha_1 + b, \alpha_2 + b$ 线性相关, $\therefore$ 存在不全为零的数

x_1, x_2 , 使得 $x_1(\alpha_1 + b) + x_2(\alpha_2 + b) = 0$,

即 $x_1\alpha_1 + x_2\alpha_2 = -(x_1 + x_2)b$, 而 $x_1 + x_2 \neq 0$,

如若不然, 有 $x_1\alpha_1 + x_2\alpha_2 = 0$, 则 α_1, α_2 线性相关, 与已知

α_1, α_2 线性无关矛盾. 于是

$$b = -\frac{x_1}{x_1 + x_2}\alpha_1 - \frac{x_2}{x_1 + x_2}\alpha_2 = c\alpha_1 + (-1-c)\alpha_2, (c \in R).$$

(解二) 设 $b = k_1\alpha_1 + k_2\alpha_2, \therefore \alpha_1 + b, \alpha_2 + b$ 线性相关

$$\therefore 0 = (\alpha_1 + b, \alpha_2 + b) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = [(\alpha_1, \alpha_2) + (b, b)] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = [(\alpha_1, \alpha_2) + (\alpha_1, \alpha_2) \begin{pmatrix} k_1 & k_1 \\ k_2 & k_2 \end{pmatrix}] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ = (\alpha_1, \alpha_2) \begin{pmatrix} 1+k_1 & k_1 \\ k_2 & 1+k_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ 有非零解,}$$

$$\therefore \alpha_1, \alpha_2 \text{ 线性无关, } \therefore \begin{pmatrix} 1+k_1 & k_1 \\ k_2 & 1+k_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \text{ 有非零解, } \therefore \begin{vmatrix} 1+k_1 & k_1 \\ k_2 & 1+k_2 \end{vmatrix} = 0,$$

$$\therefore k_1 + k_2 = -1, \quad b = c\alpha_1 + (-1-c)\alpha_2, (c \in R).$$

7. 设 a_1, a_2 线性相关, b_1, b_2 也线性相关,问 $a_1 + b_1, a_2 + b_2$ 是否一定线性相关? 试举例说明之

解 不一定.例: $a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, a_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, b_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, b_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$ 满足 a_1, a_2 线性相关, b_1, b_2 也线性相关,但

$a_1 + b_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, a_2 + b_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ 线性无关.

8. 举例说明下列命题是错误的:

(1) 若向量组 $a_1, a_2, \dots, a_m$ 是线性相关的,则 a_1 可由 $a_2, \dots, a_m$ 线性表示.

(2) 若有不全为0的数 $\lambda_1, \lambda_2, \dots, \lambda_m$,使 $\lambda_1 a_1 + \dots + \lambda_m a_m + \lambda_1 b_1 + \dots + \lambda_m b_m = 0$ 成立,则 $a_1, \dots, a_m$ 线性相关, $b_1, \dots, b_m$ 亦线性相关.

(3) 若只有当 $\lambda_1, \dots, \lambda_m$ 全为0时,等式 $\lambda_1 a_1 + \dots + \lambda_m a_m + \lambda_1 b_1 + \dots + \lambda_m b_m = 0$ 才能成立,则 $a_1, \dots, a_m$ 线性无关, $b_1, \dots, b_m$ 亦线性无关.

(4) 若 $a_1, \dots, a_m$ 线性相关, $b_1, \dots, b_m$ 亦线性相关,则有不全为0的数 $\lambda_1, \lambda_2, \dots, \lambda_m$,使 $\lambda_1 a_1 + \dots + \lambda_m a_m = 0, \lambda_1 b_1 + \dots + \lambda_m b_m = 0$ 同时成立.

解 (1) $a_1 = (1, 0, 0, \dots, 0)^T, a_2 = a_3 = \dots = a_m = 0$, 满足 $a_1, a_2, \dots, a_m$ 线性相关,但 a_1 不能由 $a_2, \dots, a_m$ 线性表示.

(2) 有不全为0的数 $\lambda_1, \lambda_2, \dots, \lambda_m$,使 $\lambda_1 a_1 + \dots + \lambda_m a_m + \lambda_1 b_1 + \dots + \lambda_m b_m = 0$

$\Rightarrow \lambda_1 (a_1 + b_1) + \dots + \lambda_m (a_m + b_m) = 0 \Rightarrow a_1 + b_1, \dots, a_m + b_m$ 线性相关,

令 $a_1 = -b_1 = e_1, \dots, a_m = -b_m = e_m$, 则 $a_1, \dots, a_m$ 线性无关, $b_1, \dots, b_m$ 亦线性无关,

$a_1 + b_1 = 0, \dots, a_m + b_m = 0$ 线性相关.

(3) 只有当 $\lambda_1, \dots, \lambda_m$ 全为0时,等式 $\lambda_1 a_1 + \dots + \lambda_m a_m + \lambda_1 b_1 + \dots + \lambda_m b_m = 0$,

即等式 $\lambda_1 (a_1 + b_1) + \dots + \lambda_m (a_m + b_m) = 0$ 才能成立 $\Rightarrow a_1 + b_1, \dots, a_m + b_m$ 线性无关,

令 $a_1 = a_2 = \dots = a_m = 0$ 线性相关, $b_1 = e_1, b_2 = e_2, \dots, b_m = e_m$ 线性无关,

$a_1 + b_1, \dots, a_m + b_m$ 线性无关.

(4) $a_1 = (1, 0)^T, a_2 = (2, 0)^T$ 线性相关, $b_1 = (0, 3)^T, b_2 = (0, 4)^T$ 线性相关,

$\begin{cases} \lambda_1 a_1 + \lambda_2 a_2 = 0 \\ \lambda_1 b_1 + \lambda_2 b_2 = 0 \end{cases} \Rightarrow \lambda_1 = \lambda_2 = 0$ 与题设矛盾.

9. 设 $b_1 = a_1 + a_2, b_2 = a_2 + a_3, b_3 = a_3 + a_4, b_4 = a_4 + a_1$,

证明向量组 b_1, b_2, b_3, b_4 线性相关.

(证一) 要证向量组 b_1, b_2, b_3, b_4 线性相关,只需线性方程组

$$0 = (b_1, b_2, b_3, b_4) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = (a_1, a_2, a_3, a_4) \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \quad (1)$$

有非零解,

$$\text{令 } \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 0 \quad (2) \text{ 由于 } \begin{vmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} = 0, \text{ 故线性方程组(2)有非零解,}$$

从而线性方程组(1)有非零解, 得证向量组 b_1, b_2, b_3, b_4 线性相关.

(证二) 设 $x_1 b_1 + x_2 b_2 + x_3 b_3 + x_4 b_4 = 0$, 则

$$x_1 (a_1 + a_2) + x_2 (a_2 + a_3) + x_3 (a_3 + a_4) + x_4 (a_4 + a_1) = 0$$

即

$$(x_1 + x_4) a_1 + (x_1 + x_2) a_2 + (x_2 + x_3) a_3 + (x_3 + x_4) a_4 = 0 \quad (*)$$

(1) 若 a_1, a_2, a_3, a_4 线性相关,

则存在不全为零的数 k_1, k_2, k_3, k_4 ,

$$k_1 = x_1 + x_4, k_2 = x_1 + x_2, k_3 = x_2 + x_3, k_4 = x_3 + x_4,$$

使得(*)成立, 知 x_1, x_2, x_3, x_4 不全为零, 得证向量组 b_1, b_2, b_3, b_4 线性相关;

$$(2) \text{ 若 } a_1, a_2, a_3, a_4 \text{ 线性无关, 则 } \begin{cases} x_1 + x_4 = 0 \\ x_1 + x_2 = 0 \\ x_2 + x_3 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 0 \text{ 有非零解, 得证向量组 } b_1, b_2, b_3, b_4 \text{ 线性相关.}$$

$$(\text{证三}) (b_1, b_2, b_3, b_4) = (a_1, a_2, a_3, a_4) \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix},$$

$$R(b_1, b_2, b_3, b_4) \leq R \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = 3 < 4, \therefore \text{向量组 } b_1, b_2, b_3, b_4 \text{ 线性相关.}$$

(证四) $\because b_1 - b_2 + b_3 - b_4 = 0, \therefore$ 向量组 b_1, b_2, b_3, b_4 线性相关.

10. 设 $b_1 = a_1, b_2 = a_1 + a_2, \dots, b_r = a_1 + a_2 + \dots + a_r$, 且向量组 $a_1, a_2, \dots, a_r$ 线性无关, 证明向量组 $b_1, b_2, \dots, b_r$ 线性无关.

证明 设有数 $x_1, x_2, \dots, x_r$, 使得 $x_1 b_1 + x_2 b_2 + \dots + x_r b_r = 0$, 则

$$x_1 a_1 + x_2 (a_1 + a_2) + \dots + x_r (a_1 + a_2 + \dots + a_r) = 0$$

$$(x_1 + x_2 + \dots + x_r) a_1 + (x_2 + \dots + x_r) a_2 + \dots + x_r a_r = 0$$

$$\because \text{向量组 } a_1, a_2, \dots, a_r \text{ 线性无关, } \therefore \begin{cases} x_1 + x_2 + \dots + x_r = 0 \\ x_2 + \dots + x_r = 0 \\ \dots \\ x_r = 0 \end{cases} \Rightarrow x_1 = x_2 = \dots = x_r = 0,$$

$\therefore$ 向量组 $b_1, b_2, \dots, b_r$ 线性无关

11. 求下列向量组的秩, 并求一个最大无关组:

$$(1) a_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \\ 4 \end{pmatrix}, a_2 = \begin{pmatrix} 9 \\ 100 \\ 10 \\ 4 \end{pmatrix}, a_3 = \begin{pmatrix} -2 \\ -4 \\ 2 \\ -8 \end{pmatrix}; \quad (2) a_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}, a_2 = \begin{pmatrix} 4 \\ -1 \\ -5 \\ -6 \end{pmatrix}, a_3 = \begin{pmatrix} 1 \\ -3 \\ -4 \\ -7 \end{pmatrix}.$$

$$\text{解: } (1) (a_1, a_2, a_3) = \begin{pmatrix} 1 & 9 & -2 \\ 2 & 100 & -4 \\ -1 & 10 & 2 \\ 4 & 4 & -8 \end{pmatrix} \sim \begin{pmatrix} 1 & 9 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{秩为2, 一组最大线性无关组为 } a_1, a_2.$$

$$(2) (a_1, a_2, a_3) = \begin{pmatrix} 1 & 4 & 1 \\ 2 & -1 & -3 \\ 1 & -5 & -4 \\ 3 & -6 & -7 \end{pmatrix} \sim \begin{pmatrix} 1 & 4 & 1 \\ 0 & -9 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{秩为2, 一组最大线性无关组为 } a_1, a_2.$$

12. 利用初等行变换求下列矩阵的列向量组的一个最大无关组, 并把其余列向量用最大无关组线性表示:

$$(1) \begin{pmatrix} 25 & 31 & 17 & 43 \\ 75 & 94 & 53 & 132 \\ 75 & 94 & 54 & 134 \\ 25 & 32 & 20 & 48 \end{pmatrix}; \quad (2) \begin{pmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 2 & 1 & 5 & -1 \\ 2 & 0 & 3 & -1 & 3 \\ 1 & 1 & 0 & 4 & -1 \end{pmatrix}.$$

$$\text{解: } (1) \begin{pmatrix} 25 & 31 & 17 & 43 \\ 75 & 94 & 53 & 132 \\ 75 & 94 & 54 & 134 \\ 25 & 32 & 20 & 48 \end{pmatrix} \sim \begin{pmatrix} 25 & 31 & 17 & 43 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & \frac{8}{5} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \therefore \text{秩为3,}$$

$$\therefore (a_1, a_2, a_3) \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{知 } R(a_1, a_2, a_3) = 3, \text{故 } a_1, a_2, a_3 \text{ 线性无关.}$$

$$\therefore \text{最大无关组为 } a_1, a_2, a_3. \quad a_4 = \frac{8}{5}a_1 - a_2 + 2a_3.$$

$$(2) \begin{pmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 2 & 1 & 5 & -1 \\ 2 & 0 & 3 & -1 & 3 \\ 1 & 1 & 0 & 4 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \therefore \text{秩为3}$$

$$\therefore (a_1, a_2, a_3) \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{知 } R(a_1, a_2, a_3) = 3, \text{故 } a_1, a_2, a_3 \text{ 线性无关.}$$

$$\therefore \text{最大无关组为 } a_1, a_2, a_3. \quad a_4 = a_1 + 3a_2 - a_3, \quad a_5 = -a_2 + a_3.$$

$$13. \text{设向量组 } \begin{pmatrix} a \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ b \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \text{ 的秩为2, 求 } a, b.$$

$$\text{解 } \begin{pmatrix} 1 & 2 & a & 2 \\ 2 & 3 & 3 & b \\ 1 & 1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & a & 2 \\ 0 & -1 & 3-2a & b-4 \\ 0 & 0 & a-2 & 3-b \end{pmatrix},$$

$$\therefore \text{秩为2, } \therefore a-2=0, 3-b=0, \therefore a=2, b=3.$$

14. 设 $a_1, a_2, \dots, a_n$ 是一组 n 维向量, 已知 n 维单位坐标向量组 $e_1, e_2, \dots, e_n$ 能由它们线性表示, 证明 $a_1, a_2, \dots, a_n$ 线性无关.

$$\text{证明 } \because n = R(e_1, e_2, \dots, e_n) \leq R(a_1, a_2, \dots, a_n) \leq n, \therefore R(a_1, a_2, \dots, a_n) = n,$$

$$\therefore a_1, a_2, \dots, a_n \text{ 线性无关.}$$

15. 设 $a_1, a_2, \dots, a_n$ 是一组 n 维向量, 证明它们线性无关

的充分必要条件是: 任一 n 维向量都可由它们线性表示.

$$\text{证明 "} \Rightarrow \text{" } \because a_1, a_2, \dots, a_n \text{ 线性无关, } R^n \text{ 的秩为 } n,$$

$$\therefore a_1, a_2, \dots, a_n \text{ 是 } R^n \text{ 的一个最大无关组,}$$

$$\therefore \text{任一 } n \text{ 维向量都可由它们线性表示.}$$

$$\text{"} \Leftarrow \text{" 任一 } n \text{ 维向量都可由它们线性表示, } n \text{ 维单位坐标向量组 } e_1, e_2, \dots, e_n \text{ 能由它们线性表示,}$$

$$\therefore a_1, a_2, \dots, a_n \text{ 线性无关.}$$

16. 设向量组 $a_1, a_2, \dots, a_m$ 线性相关, 且 $a_1 \neq 0$, 证明存在某个向量 $a_k (2 \leq k \leq m)$,

使 a_k 能由 $a_1, \dots, a_{k-1}$ 线性表示.

证明 $\because a_1, a_2, \dots, a_m$ 线性相关, $\therefore$ 存在不全为零的数 $x_1, x_2, \dots, x_m$, 使得 $x_1 a_1 + x_2 a_2 + \dots + x_m a_m = 0$

$\therefore x_2, \dots, x_m$ 中至少有一个不为零, 否则 $x_1 a_1 = 0$, 又 $a_1 \neq 0$, 有 $x_1 = 0$, 导致矛盾.

取 $k (2 \leq k \leq m)$, 满足 $x_k \neq 0, x_{k+1} = \dots = x_m = 0$, 则 $x_1 a_1 + x_2 a_2 + \dots + x_k a_k = 0$,

于是 $a_k = -\frac{x_1}{x_k} a_1 - \frac{x_2}{x_k} a_2 - \dots - \frac{x_{k-1}}{x_k} a_{k-1}$, $\therefore$ 存在向量 $a_k (2 \leq k \leq m)$, 使 a_k 能由 $a_1, \dots, a_{k-1}$ 线性表示.

17. 设向量组 $B: b_1, \dots, b_r$ 能由向量组 $A: a_1, \dots, a_r$ 线性表示为 $(b_1, \dots, b_r) = (a_1, \dots, a_r)K$

其中 K 为 $s \times r$ 矩阵, 且 A 组线性无关. 证明 B 组线性无关的充分必要条件是矩阵 K 的秩 $R(K) = r$.

证明 B 组线性无关 $\Leftrightarrow (b_1, \dots, b_r) \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} = (a_1, \dots, a_r)K \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} = 0$ 只有零解

$\Leftrightarrow K \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} = 0$ 只有零解 $\Leftrightarrow R(K) = r$

18. 设 $\begin{cases} \beta_1 = \alpha_1 + \alpha_2 + \dots + \alpha_n, \\ \beta_2 = \alpha_1 + \alpha_2 + \dots + \alpha_{n-1}, \\ \dots \\ \beta_n = \alpha_1 + \alpha_2 + \dots + \alpha_{n-1}. \end{cases}$ 证明向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 与向量组 $\beta_1, \beta_2, \dots, \beta_n$ 等价.

证明 $(\beta_1, \beta_2, \dots, \beta_n) = (\alpha_1, \alpha_2, \dots, \alpha_n)\Phi$, 其中 $\Phi = \begin{pmatrix} 0 & 1 & \dots & 1 \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 0 \end{pmatrix}$

$|\Phi| = (-1)^{n-1}(n-1)$, 由题意知 $n > 1$, 故 $|\Phi| \neq 0$, Φ 可逆. 于是 $(\alpha_1, \alpha_2, \dots, \alpha_n) = (\beta_1, \beta_2, \dots, \beta_n)\Phi^{-1}$

因此向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 与向量组 $\beta_1, \beta_2, \dots, \beta_n$ 等价.

19. 已知 3 阶矩阵 A 与 3 维列向量 x 满足 $A^3 x = 3Ax - A^2 x$, 且向量组 $x, Ax, A^2 x$ 线性无关.

(1) 记 $P = (x, Ax, A^2 x)$, 求 3 阶矩阵 B , 使 $AP = PB$; (2) 求 $|A|$.

解 (1) $\because AP = A(x, Ax, A^2 x) = (Ax, A^2 x, A^3 x) = (x, Ax, A^2 x) \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix}$, $\therefore B = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix}$

(2) $\because A^3 x = 3Ax - A^2 x, \therefore A(A^2 x - 3x + Ax) = 0$, 而 $A^2 x - 3x + Ax \neq 0$, 否则 $x, Ax, A^2 x$ 线性相关, 与已知矛盾.

即 $AY = 0$ 有非零解, $Y = A^2 x - 3x + Ax$, 于是 $|A| = 0$.

20. 求下列齐次线性方程组的基础解系:

(1) $\begin{cases} x_1 - 8x_2 + 10x_3 + 2x_4 = 0 \\ 2x_1 + 4x_2 + 5x_3 - x_4 = 0 \\ 3x_1 + 8x_2 + 6x_3 - 2x_4 = 0 \end{cases}$ (2) $\begin{cases} 2x_1 - 3x_2 - 2x_3 + x_4 = 0 \\ 3x_1 + 5x_2 + 4x_3 - 2x_4 = 0 \\ 8x_1 + 7x_2 + 6x_3 - 3x_4 = 0 \end{cases}$ (3) $nx_1 + (n-1)x_2 + \dots + 2x_{n-1} + x_n = 0$

解: (1) $A = \begin{pmatrix} 1 & -8 & 10 & 2 \\ 2 & 4 & 5 & -1 \\ 3 & 8 & 6 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 4 & 0 \\ 0 & 4 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $R(A) = 2$,

基础解系含两个向量,可取为 $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} -4 \\ 0 \\ 1 \\ -3 \end{pmatrix}$.

$$(2) A = \begin{pmatrix} 2 & -3 & -2 & 1 \\ 3 & 5 & 4 & -2 \\ 8 & 7 & 6 & -3 \end{pmatrix} \sim \begin{pmatrix} -19 & 0 & -2 & 1 \\ -7 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, R(A) = 2,$$

基础解系含两个向量,可取为 $\begin{pmatrix} 1 \\ 7 \\ 0 \\ 19 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}$.

$$(3) A = (n \ n-1 \ \cdots \ 2 \ 1), R(A) = 1,$$

基础解系含 $n-1$ 个向量,可取为 $\begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \\ n \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ \vdots \\ 0 \\ n-1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ -1 \\ 2 \end{pmatrix}$.

$$21. \text{ 设 } A = \begin{pmatrix} 2 & -2 & 1 & 3 \\ 9 & -5 & 2 & 8 \end{pmatrix}, \text{ 求一个 } 4 \times 2 \text{ 矩阵 } B, \text{ 使 } AB = 0, \text{ 且 } R(B) = 2.$$

$$\text{解 } \because A = \begin{pmatrix} 2 & -2 & 1 & 3 \\ 9 & -5 & 2 & 8 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -\frac{1}{8} & \frac{1}{8} \\ 0 & 1 & -\frac{5}{8} & -\frac{11}{8} \end{pmatrix},$$

$$\therefore Ax = 0 \text{ 的基础解系为 } b_1 = \begin{pmatrix} 1 \\ 5 \\ 8 \\ 0 \end{pmatrix}, b_2 = \begin{pmatrix} -1 \\ 11 \\ 0 \\ 8 \end{pmatrix}$$

$$\text{令 } B = \begin{pmatrix} 1 & -1 \\ 5 & 11 \\ 8 & 0 \\ 0 & 8 \end{pmatrix} \text{ 满足条件 } R(B) = 2, AB = 0. \text{ (满足条件的 } B \text{ 不唯一)}$$

$$22. \text{ 求一个齐次线性方程组,使它的基础解系为 } \xi_1 = (0, 1, 2, 3)^T, \xi_2 = (3, 2, 1, 0)^T.$$

(分析: 由题意知欲求 $Ax = 0$ 使其满足: $A\xi_1 = 0, A\xi_2 = 0, R(A) = 4 - 2 = 2$. 由 $\xi_1^T A^T = 0, \xi_2^T A^T = 0$,

$$\text{得 } \begin{pmatrix} \xi_1^T \\ \xi_2^T \end{pmatrix} A^T = 0,$$

故问题转化为求 $\begin{pmatrix} \xi_1^T \\ \xi_2^T \end{pmatrix} x = 0$ 的基础解系.)

$$\text{解 } \begin{pmatrix} \xi_1^T \\ \xi_2^T \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \end{pmatrix}, \text{ 基础解系可取为 } \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{令 } A^T = \begin{pmatrix} 1 & 2 \\ -2 & -3 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, A = \begin{pmatrix} 1 & -2 & 1 & 0 \\ 2 & -3 & 0 & 1 \end{pmatrix} \text{ 则 } Ax = 0, \text{ 即齐次线性方程组 } \begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_1 - 3x_2 + x_4 = 0 \end{cases} \text{ 为所求.}$$

(A 不唯一)

23. 设四元齐次线性方程组

$$\text{I } \begin{cases} x_1 + x_2 = 0 \\ x_2 - x_4 = 0 \end{cases} \quad \text{II } \begin{cases} x_1 - x_2 + x_3 = 0 \\ x_2 - x_3 + x_4 = 0 \end{cases}$$

求: (1) 方程组 I 与 II 的基础解系; (2) I 与 II 的公共解.

解: (1) $\therefore \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix}$, $\therefore$ 方程组 I 的基础解系为 $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

$\therefore \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{pmatrix}$, $\therefore$ 方程组 II 的基础解系为 $\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix}$

$$x = c_1 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, (c_1, c_2 \in \mathbb{R})$$

(2) 方程组 I 的通解为

$$x = k_1 \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix}, (k_1, k_2 \in \mathbb{R})$$

方程组 II 的通解为

$$\text{令 } c_1 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix} = k_1 \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -c_2 \\ c_2 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -k_2 \\ k_1 - k_2 \\ k_1 \\ k_2 \end{pmatrix} \Rightarrow \begin{cases} c_1 = k_1 \\ c_2 = k_2 \\ c_2 = k_1 - k_2 \end{cases} \Rightarrow \begin{cases} c_1 = k_1 = 2c \\ c_2 = k_2 = c \end{cases}, (c \in \mathbb{R})$$

$$x = c \begin{pmatrix} -1 \\ 1 \\ 2 \\ 1 \end{pmatrix}, (c \in \mathbb{R})$$

所以 I 与 II 的公共解为:

24. 设 n 阶矩阵 A 满足 $A^2 = A$, E 为 n 阶单位阵, 证明 $R(A) + R(A-E) = n$.

证明 $A^2 = A \Rightarrow A(A-E) = 0 \Rightarrow R(A) + R(A-E) \leq n$ (矩阵秩的性质 8)

$R(A) + R(A-E) = R(A) + R(E-A) \geq R(A+E-A) = R(E) = n$ (矩阵秩的性质 6)

所以 $R(A) + R(A-E) = n$.

25. 设 A 为 n 阶矩阵 ($n \geq 2$), A^* 为 A 的伴随矩阵, 证明 $R(A^*) = \begin{cases} n, & \text{当 } R(A) = n, \\ 1, & \text{当 } R(A) = n-1, \\ 0, & \text{当 } R(A) \leq n-2. \end{cases}$

证明: 当 $R(A) = n$ 时, A 可逆, 从而 A^* 可逆, 故 $R(A^*) = n$;

当 $R(A) < n$ 时, $\therefore AA^* = |A|E = 0 \cdot E = 0$, $\therefore R(A^*) + R(A) \leq n$,

当 $R(A) = n-1$ 时, A 中至少存在一个 $n-1$ 阶子式不为 0, 从而 $A^* \neq 0$, 至少有一个非零元素, 故 $R(A^*) > 0$, 所以 $R(A^*) = 1$;

当 $R(A) \leq n-2$ 时, A 中所有 $n-1$ 阶子式全为 0, 从而 $A^* = 0$, 所以 $R(A^*) = 0$.

26. 求下列非齐次方程组的一个解及对应的齐次方程组的基础解系:

$$(1) \begin{cases} x_1 + x_2 = 5 \\ 2x_1 + x_2 + x_3 + 2x_4 = 1 \\ 5x_1 + 3x_2 + 2x_3 + 2x_4 = 3 \end{cases} \quad (2) \begin{cases} x_1 - 5x_2 + 2x_3 - 3x_4 = 11 \\ 5x_1 + 3x_2 + 6x_3 - x_4 = -1 \\ 2x_1 + 4x_2 + 2x_3 + x_4 = -6 \end{cases}$$

解: (1) $\therefore \begin{pmatrix} 1 & 1 & 0 & 0 & 5 \\ 2 & 1 & 1 & 2 & 1 \\ 5 & 3 & 2 & 2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 0 & -8 \\ 0 & 1 & -1 & 0 & 13 \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix}$,

$$\therefore \begin{cases} x_1 = -x_3 - 8 \\ x_2 = x_3 + 13 \\ x_4 = 2 \end{cases} (*)$$

$\therefore$ 令 $x_3 = 0$, 得 $\eta^* = \begin{pmatrix} -8 \\ 13 \\ 0 \\ 2 \end{pmatrix}$ 是该方程组的一个特解;

由(*)对应的齐次线性方程组 $\begin{cases} x_1 = -x_3 \\ x_2 = x_3 \\ x_4 = 0 \end{cases}$, 即得所对应的

齐次线性方程组的基础解系 $\xi = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$;

$$(2) \therefore \begin{pmatrix} 1 & -5 & 2 & -3 & 11 \\ 5 & 3 & 6 & -1 & -1 \\ 2 & 4 & 2 & 1 & -6 \end{pmatrix} \sim \begin{pmatrix} 1 & 9 & 0 & 4 & -17 \\ 0 & -7 & 1 & -\frac{7}{2} & 14 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\therefore \begin{cases} x_1 = -9x_2 - 4x_4 - 17 \\ x_3 = 7x_2 + \frac{7}{2}x_4 + 14 \end{cases} (*)$$

令 $x_2 = x_4 = 0$, 得 $\eta^* = \begin{pmatrix} -17 \\ 0 \\ 14 \\ 0 \end{pmatrix}$ 是该方程组的一个特解;

由(*)对应的齐次线性方程组 $\begin{cases} x_1 = -9x_2 - 4x_4 \\ x_3 = 7x_2 + \frac{7}{2}x_4 \end{cases}$, 即得所

对应的齐次线性方程组的基础解系 $\xi_1 = \begin{pmatrix} -9 \\ 1 \\ 7 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} -4 \\ 0 \\ \frac{7}{2} \\ 1 \end{pmatrix}$.

27. 设四元非齐次线性方程组的系数矩阵的秩为3, 已知 η_1, η_2, η_3 是它的三个解向量, 且

$$\eta_1 = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \quad \eta_2 + \eta_3 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

求该方程组的通解.

解 $\because$ 系数矩阵的秩是3, $\therefore$ 所对应的齐次线性方程组的基础解系含 $4-3=1$ 个解向量,

而 $\xi = \eta_2 + \eta_3 - \eta_1 - \eta_1 = \begin{pmatrix} -3 \\ -4 \\ -5 \\ -6 \end{pmatrix}$ 是所对应的齐次线性方程组的一个非零解,

$\therefore x = k\xi + \eta_1 (k \in R)$ 为该方程组的通解.

$$28. \text{ 设有向量组 } A: a_1 = \begin{pmatrix} \alpha \\ 2 \\ 10 \end{pmatrix}, a_2 = \begin{pmatrix} -2 \\ 1 \\ 5 \end{pmatrix}, a_3 = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}, \text{ 及向量 } b = \begin{pmatrix} 1 \\ \beta \\ -1 \end{pmatrix},$$

问 α, β 为何值时:

(1) 向量 b 不能由向量组 A 线性表示;

(2) 向量 b 能由向量组 A 线性表示, 且表示式唯一;

(3) 向量 b 能由向量组 A 线性表示, 且表示式不唯一;

并求一般表示式.

$$\text{解: 记 } A = (a_3 \ a_2 \ a_1) = \begin{pmatrix} -1 & -2 & \alpha \\ 1 & 1 & 2 \\ 4 & 5 & 10 \end{pmatrix}, B = (a_3 \ a_2 \ a_1 \ b) = \begin{pmatrix} -1 & -2 & \alpha & 1 \\ 1 & 1 & 2 & \beta \\ 4 & 5 & 10 & -1 \end{pmatrix}$$

$$B = \begin{pmatrix} -1 & -2 & \alpha & 1 \\ 1 & 1 & 2 & \beta \\ 4 & 5 & 10 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -\alpha & -1 \\ 0 & 1 & -2-\alpha & -\beta-1 \\ 0 & 0 & 4+\alpha & -3\beta \end{pmatrix}$$

(1) 当 $\alpha = -4, \beta \neq 0$ 时, $R(A) \neq R(B)$, $\therefore$ 向量 b 不能由向量组 A 线性表示;

(2) 当 $\alpha \neq -4$ 时, $R(A) = R(B) = 3$, $\therefore$ 向量 b 能由向量组 A 线性表示, 且表示式唯一;

(3) 当 $\alpha = -4, \beta = 0$ 时, $R(A) = R(B) = 2 < 3$, 向量 b 能由向量组 A 线性表示, 且表示式不唯一, 此时由

$$B \sim \begin{pmatrix} 1 & 2 & -\alpha & -1 \\ 0 & 1 & -2-\alpha & -\beta-1 \\ 0 & 0 & 4+\alpha & -3\beta \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \text{得 } Ax = b \text{ 的通解为}$$

$$\begin{cases} x_1 = 1 \\ x_2 = -2c - 1, \therefore b = ca_1 + (-2c - 1)a_2 + a_3, (c \in R) \\ x_3 = c \end{cases}$$

29. 设 $a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, c = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$, 证明三直线 $\begin{cases} l_1: a_1x + b_1y + c_1 = 0 \\ l_2: a_2x + b_2y + c_2 = 0, (a_i^2 + b_i^2 \neq 0) \\ l_3: a_3x + b_3y + c_3 = 0 \end{cases}$

相交于一点的充要条件为: 向量组 a, b 线性无关, 且向量组 a, b, c 线性相关.

证明 三直线相交于一点 $\Leftrightarrow \begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \\ a_3x + b_3y + c_3 = 0 \end{cases}$ 有唯一解 $\Leftrightarrow (a, b) \begin{pmatrix} x \\ y \end{pmatrix} = c$ 有唯一解

$$\Leftrightarrow R(a, b) = R(a, b, c) = 2 \Leftrightarrow \text{向量组 } a, b \text{ 线性无关, 且向量组 } a, b, c \text{ 线性相关}$$

30. 设矩阵 $A = (a_1, a_2, a_3, a_4)$, 其中 a_2, a_3, a_4 线性无关, $a_1 = 2a_2 - a_3$. 向量 $b = a_1 + a_2 + a_3 + a_4$, 求 $Ax = b$ 的通解.

解 $\because a_2, a_3, a_4$ 线性无关, $a_1 = 2a_2 - a_3, \therefore R(A) = 3, \therefore Ax = 0$ 的基础解系含一个解向量,

$$\because a_1 = 2a_2 - a_3, \therefore \xi = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} \text{ 是 } Ax = 0 \text{ 的基础解系,}$$

$$\because b = a_1 + a_2 + a_3 + a_4, \therefore \eta = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \text{ 是方程 } Ax = b \text{ 的一个特解,}$$

$\therefore x = c\xi + \eta, (c \in R)$ 是 $Ax = b$ 的通解.

37. 验证 $\alpha_1 = (1, -1, 0)^T, \alpha_2 = (2, 1, 3)^T, \alpha_3 = (3, 1, 2)^T$ 为 R^3 的一个基, 并把 $\nu_1 = (5, 0, 7)^T, \nu_2 = (-9, -8, -13)^T$ 用这个基线性表示.

证明 $\because (a_1, a_2, a_3) = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 1 & 1 \\ 0 & 3 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\therefore$ 向量组 a_1, a_2, a_3 线性无关, 是 R^3 的一个基

$$\because (a_1, a_2, a_3, v_1, v_2) = \begin{pmatrix} 1 & 2 & 3 & 5 & -9 \\ -1 & 1 & 1 & 0 & -8 \\ 0 & 3 & 2 & 7 & -13 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & 3 & -3 \\ 0 & 0 & 1 & -1 & -2 \end{pmatrix}$$

$$\therefore v_1 = 2a_1 + 3a_2 - a_3, v_2 = 3a_1 - 3a_2 - 2a_3.$$

38. 已知 R^3 的两个基为

$$a_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, a_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, a_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ 及 } b_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, b_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, b_3 = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}$$

求由基 a_1, a_2, a_3 到 b_1, b_2, b_3 的过渡矩阵 P .

$$\text{解 } \because (b_1, b_2, b_3) = (a_1, a_2, a_3)P, \therefore P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & -1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 4 & 3 \end{pmatrix}$$

$$\because \begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 0 & 2 & 3 & 4 \\ 1 & -1 & 1 & 1 & 4 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 2 & 3 & 4 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \end{pmatrix}, \therefore P = \begin{pmatrix} 2 & 3 & 4 \\ 0 & -1 & 0 \\ -1 & 0 & -1 \end{pmatrix}$$

31. 设 η^* 是非齐次线性方程组的一个解, $\xi_1, \dots, \xi_{n-r}$ 是对应的齐次线性方程组的一个基础解系.

证明:

(1) $\eta^*, \xi_1, \dots, \xi_{n-r}$ 线性无关;

(2) $\eta^*, \eta^* + \xi_1, \dots, \eta^* + \xi_{n-r}$ 线性无关.

证明

(1) 设有数 $k, k_1, \dots, k_{n-r}$ 使得 $k\eta^* + k_1\xi_1 + \dots + k_{n-r}\xi_{n-r} = 0$, 则

$$A(k\eta^* + k_1\xi_1 + \dots + k_{n-r}\xi_{n-r}) = k(A\eta^*) + k_1(A\xi_1) + \dots + k_{n-r}(A\xi_{n-r}) = kb = 0,$$

$$\because b \neq 0, \therefore k = 0, \therefore k_1\xi_1 + \dots + k_{n-r}\xi_{n-r} = 0,$$

又 $\xi_1, \dots, \xi_{n-r}$ 线性无关, $\therefore k_1 = \dots = k_{n-r} = 0$, 因此 $\eta^*, \xi_1, \dots, \xi_{n-r}$ 线性无关.

$$(2) \because (\eta^*, \eta^* + \xi_1, \dots, \eta^* + \xi_{n-r}) = (\eta^*, \xi_1, \dots, \xi_{n-r})\Phi, \text{ 其中 } \Phi = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}, \text{ 而 } \Phi \text{ 可逆,}$$

$$\text{于是 } R(\eta^*, \eta^* + \xi_1, \dots, \eta^* + \xi_{n-r}) = R(\eta^*, \xi_1, \dots, \xi_{n-r}) = n - r + 1,$$

因此 $\eta^*, \eta^* + \xi_1, \dots, \eta^* + \xi_{n-r}$ 线性无关.

32. 设 $\eta_1, \dots, \eta_s$ 是非齐次线性方程组 $Ax = b$ 的 s 个解, $k_1, \dots, k_s$ 为实数, 满足 $k_1 + k_2 + \dots + k_s = 1$.

证明 $x = k_1\eta_1 + k_2\eta_2 + \dots + k_s\eta_s$ 也是它的解.

证明 $\because A(k_1\eta_1 + k_2\eta_2 + \dots + k_s\eta_s) = k_1(A\eta_1) + k_2(A\eta_2) + \dots + k_s(A\eta_s) = k_1b + k_2b + \dots + k_sb$

$= (k_1 + k_2 + \dots + k_s)b = b$,

$\therefore x = k_1\eta_1 + k_2\eta_2 + \dots + k_s\eta_s$ 也是 $Ax = b$ 的解

33. 设非齐次线性方程组 $Ax = b$ 的系数矩阵的秩为 r , $\eta_1, \dots, \eta_{n-r+1}$

是它的 $n-r+1$ 个线性无关的解.

试证它的任一解可表示为 $x = k_1\eta_1 + \dots + k_{n-r+1}\eta_{n-r+1}$ (其中 $k_1 + \dots + k_{n-r+1} = 1$).

证明 $\because \eta_1, \dots, \eta_{n-r+1}$ 是 $Ax = b$ 的解, $\therefore \xi_1 = \eta_2 - \eta_1, \dots, \xi_{n-r} = \eta_{n-r+1} - \eta_1$ 是 $Ax = 0$ 的解,

下证 $\xi_1, \xi_2, \dots, \xi_{n-r}$ 是 $Ax = 0$ 的基础解系:

设有数 $k_1, \dots, k_{n-r}$, 使得 $k_1\xi_1 + \dots + k_{n-r}\xi_{n-r} = 0$,

即 $k_1(\eta_2 - \eta_1) + \dots + k_{n-r}(\eta_{n-r+1} - \eta_1) = 0$, 即 $(-k_1 - \dots - k_{n-r})\eta_1 + k_1\eta_2 + \dots + k_{n-r}\eta_{n-r+1} = 0$,

$\because \eta_1, \dots, \eta_{n-r+1}$ 线性无关, $\therefore -k_1 - \dots - k_{n-r} = 0, k_1 = \dots = k_{n-r} = 0$

$\therefore \xi_1, \xi_2, \dots, \xi_{n-r}$ 线性无关, 是 $Ax = 0$ 的基础解系,

$\therefore x = k_2\xi_1 + k_3\xi_2 + \dots + k_{n-r}\xi_{n-r} + \eta_1, (k_2, \dots, k_{n-r} \in R)$

$= (1 - k_2 - k_3 - \dots - k_{n-r})\eta_1 + k_2\eta_2 + \dots + k_{n-r}\eta_{n-r+1} = k_1\eta_1 + k_2\eta_2 + \dots + k_{n-r+1}\eta_{n-r+1}$,

(其中 $k_1 + \dots + k_{n-r+1} = 1$)

是 $Ax = b$ 的通解.

34. 设 $V_1 = \left\{ x = (x_1, x_2, \dots, x_n)^T \mid \begin{array}{l} x_1, \dots, x_n \in R \\ \text{满足 } x_1 + \dots + x_n = 0 \end{array} \right\}, V_2 = \left\{ x = (x_1, x_2, \dots, x_n)^T \mid \begin{array}{l} x_1, \dots, x_n \in R \\ \text{满足 } x_1 + \dots + x_n = 1 \end{array} \right\}$,

问 V_1, V_2 是不是向量空间? 为什么?

答 V_1 是向量空间, 因为

(1) V_1 非空; (2) 若 $a = (a_1, a_2, \dots, a_n)^T \in V_1, b = (b_1, b_2, \dots, b_n)^T \in V_1$,

$\therefore (a_1 + b_1) + (a_2 + b_2) + \dots + (a_n + b_n) = (a_1 + a_2 + \dots + a_n) + (b_1 + b_2 + \dots + b_n) = 0$

$\therefore a + b = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)^T \in V_1$,

(3) $\forall \lambda \in R, \therefore \lambda a_1 + \lambda a_2 + \dots + \lambda a_n = 0, \therefore \lambda a = (\lambda a_1, \lambda a_2, \dots, \lambda a_n)^T \in V_1$.

V_2 不是向量空间, 因为若 $a = (a_1, a_2, \dots, a_n)^T \in V_2$, 因 $2a_1 + 2a_2 + \dots + 2a_n = 2$,

故 $2a = (2a_1, 2a_2, \dots, 2a_n)^T \notin V_2$

35. 试证: 由 $a_1 = (0, 1, 1)^T, a_2 = (1, 0, 1)^T, a_3 = (1, 1, 0)^T$ 所生成的向量空间就是 R^3 .

证明 $\because (a_1, a_2, a_3) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\therefore$ 向量组 a_1, a_2, a_3 线性无关, 是 R^3 的一个基, $\therefore$ 由 a_1, a_2, a_3 所生成的向量空间就是 R^3

36. 由 $a_1 = (1, 1, 0, 0)^T, a_2 = (1, 0, 1, 1)^T$ 所生成的向量空间记作 L_1 , 由 $b_1 = (2, -1, 3, 3)^T, b_2 = (0, 1, -1, -1)^T$ 所生成的

向量空间记作 L_2 , 试证 $L_1 = L_2$.

证明

$\because (a_1, a_2) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, (b_1, b_2) = \begin{pmatrix} 2 & 0 \\ -1 & 1 \\ 3 & -1 \\ 3 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\therefore (a_1, a_2) \sim (b_1, b_2), \therefore$ 向量组 a_1, a_2 与向量组 b_1, b_2 等价, 因此它们所生成的向量空间相等, 即 $L_1 = L_2$.

第五章

1. 设 $a = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$, $b = \begin{pmatrix} -4 \\ 2 \\ 3 \end{pmatrix}$, c 与 a 正交, 且 $b = \lambda a + c$, 求 λ 和 c .

解: $c = b - \lambda a = \begin{pmatrix} -4-\lambda \\ 2 \\ 3+2\lambda \end{pmatrix}$, c 与 a 正交, 有 $(-4-\lambda) \cdot 1 + 2 \cdot 0 + (3+2\lambda) \cdot (-2) = 0$, 解得 $\lambda = -2$, $c = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$

2. 试用施密特法把下列向量组正交化:

(1) $(a_1, a_2, a_3) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$; (2) $(a_1, a_2, a_3) = \begin{pmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

解 (1) 根据施密特正交方法:

$$\text{令 } b_1 = a_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, b_2 = a_2 - \frac{[a_2, b_1]}{[b_1, b_1]} b_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, b_3 = a_3 - \frac{[a_3, b_1]}{[b_1, b_1]} b_1 - \frac{[a_3, b_2]}{[b_2, b_2]} b_2 = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

故正交化后得: $(b_1, b_2, b_3) = \begin{pmatrix} 1 & -1 & \frac{1}{3} \\ 1 & 0 & -\frac{2}{3} \\ 1 & 1 & \frac{1}{3} \end{pmatrix}$

(2) 根据施密特正交方法:

$$\text{令 } b_1 = a_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \end{pmatrix}, b_2 = a_2 - \frac{[a_2, b_1]}{[b_1, b_1]} b_1 = \frac{1}{3} \begin{pmatrix} 1 \\ -3 \\ 2 \\ 1 \end{pmatrix}, b_3 = a_3 - \frac{[a_3, b_1]}{[b_1, b_1]} b_1 - \frac{[a_3, b_2]}{[b_2, b_2]} b_2 = \frac{1}{5} \begin{pmatrix} -1 \\ 3 \\ 3 \\ 4 \end{pmatrix}$$

故正交化后得: $(b_1, b_2, b_3) = \begin{pmatrix} 1 & \frac{1}{3} & -\frac{1}{5} \\ 0 & -1 & \frac{3}{5} \\ -1 & \frac{2}{3} & \frac{3}{5} \\ 1 & \frac{1}{3} & \frac{4}{5} \end{pmatrix}$

3. 下列矩阵是不是正交矩阵? 并说明理由:

$$(1) \begin{pmatrix} 1 & -\frac{1}{2} & \frac{1}{3} \\ -\frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & -1 \end{pmatrix}; \quad (2) \begin{pmatrix} \frac{1}{9} & -\frac{8}{9} & -\frac{4}{9} \\ -\frac{8}{9} & \frac{1}{9} & -\frac{4}{9} \\ -\frac{4}{9} & -\frac{4}{9} & \frac{7}{9} \end{pmatrix}$$

解: (1) 第一个列向量非单位向量, 故不是正交阵;

(2) 该方阵每一个列向量均是单位向量, 且两两正交, 故为正交阵.

4. 设 x 为 n 维列向量, $x^T x = 1$, 令 $H = E - 2xx^T$, 证明 H 是对称的正交阵.

解: $\because H^T = (E - 2xx^T)^T = E^T - 2(xx^T)^T = E - 2(x^T)^T x^T = E - 2xx^T = H$, $\therefore H$ 是对称阵;

$$\begin{aligned} \because H^T H &= (E - 2xx^T)(E - 2xx^T) = E - 2xx^T E - E 2xx^T + 2xx^T 2xx^T \\ &= E - 4xx^T + 4x(x^T x)x^T = E - 4xx^T + 4xx^T = E, \end{aligned}$$

$\therefore H$ 是正交阵.

5. 设 A, B 都是正交阵, 证明 AB 也是正交阵.

证 $\because (AB)^T(AB) = B^T(A^T A)B = B^T E B = B^T B = E, \therefore AB$ 也是正交阵.

6. 求下列矩阵的特征值与特征向量:

$$(1) \begin{pmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{pmatrix}; (2) \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}; (3) \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

$$\text{解(1)} |A - \lambda E| = \begin{vmatrix} 2-\lambda & -1 & 2 \\ 5 & -3-\lambda & 3 \\ -1 & 0 & -2-\lambda \end{vmatrix} = -(\lambda+1)^3, \text{故 } A \text{ 的特征值为 } \lambda_1 = \lambda_2 = \lambda_3 = -1$$

$$\text{当 } \lambda_1 = \lambda_2 = \lambda_3 = -1 \text{ 时, 解方程组 } (A+E)x=0, \text{ 由 } A+E = \begin{pmatrix} 2 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

得基础解系 $p = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$, 故 $k_1 p (k_1 \neq 0)$ 是对应于 $\lambda_1 = \lambda_2 = \lambda_3 = -1$ 的全部特征向量.

$$(2) |A - \lambda E| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 1-\lambda & 3 \\ 3 & 3 & 6-\lambda \end{vmatrix} = -\lambda(\lambda+1)(\lambda-9), \text{故 } A \text{ 的特征值为 } \lambda_1 = 0, \lambda_2 = -1, \lambda_3 = 9$$

$$\text{当 } \lambda_1 = 0 \text{ 时, 解方程组 } Ax=0, \text{ 由 } A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{得基础解系 } p_1 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

故 $k_1 p_1 (k_1 \neq 0)$ 是对应于 $\lambda_1 = 0$ 的全部特征向量.

$$\text{当 } \lambda_2 = -1 \text{ 时, 解方程组 } (A+E)x=0, \text{ 由 } A+E = \begin{pmatrix} 2 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{得基础解系 } p_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},$$

故 $k_2 p_2 (k_2 \neq 0)$ 是对应于 $\lambda_2 = -1$ 的全部特征向量.

$$\text{当 } \lambda_3 = 9 \text{ 时, 解方程组 } (A-9E)x=0, \text{ 由 } A-9E = \begin{pmatrix} -8 & 2 & 3 \\ 2 & -8 & 3 \\ 3 & 3 & -3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}, \text{得基础解系 } p_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix},$$

故 $k_3 p_3 (k_3 \neq 0)$ 是对应于 $\lambda_3 = 9$ 的全部特征向量.

$$(3) |A - \lambda E| = \begin{vmatrix} -\lambda & 0 & 0 & 1 \\ 0 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 0 \\ 1 & 0 & 0 & -\lambda \end{vmatrix} = (\lambda+1)^2(\lambda-1)^2, \text{故 } A \text{ 的特征值为 } \lambda_1 = \lambda_2 = -1, \lambda_3 = \lambda_4 = 1$$

$$\text{当 } \lambda_1 = \lambda_2 = -1 \text{ 时, 由 } A+E = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \text{得基础解系 } p_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, p_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

$k_1 p_1 + k_2 p_2, (k_1^2 + k_2^2 \neq 0)$ 是对应于 $\lambda_1 = \lambda_2 = -1$ 的全部特征向量.

$$\text{当 } \lambda_3 = \lambda_4 = 1 \text{ 时, 由 } A-E = \begin{pmatrix} -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \text{得基础解系 } p_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, p_4 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

$k_3 p_3 + k_4 p_4, (k_3^2 + k_4^2 \neq 0)$ 是对应于 $\lambda_3 = \lambda_4 = 1$ 的全部特征向量.

7. 设 A 为 n 阶矩阵, 证明 A^T 与 A 的特征值相同.

证 $\because |A - \lambda E| = |(A - \lambda E)^T| = |A^T - \lambda E|, \therefore A^T$ 与 A 的特征值相同.

8. 设 n 阶矩阵 A, B 满足 $R(A) + R(B) < n$, 证明 A 与 B 有公共的特征值, 有公共的特征向量.

证 $\because R \begin{pmatrix} A \\ B \end{pmatrix} \leq R(A) + R(B) < n, \therefore \begin{pmatrix} A \\ B \end{pmatrix} x = 0$ 有非零解,

$\therefore$ 有 $p \neq 0$ 使得: $Ap = 0 \cdot p, Bp = 0 \cdot p$, 即 A 与 B 有公共的特征值 0 , 有公共的特征向量 p .

9. 设 $A^2 - 3A + 2E = 0$, 证明 A 的特征值只能取 1 或 2.

证 若 λ 是 A 的特征值, 则 $\varphi(\lambda) = \lambda^2 - 3\lambda + 2$ 是 $\varphi(A) = A^2 - 3A + 2E$ 的特征值,

又 $\varphi(A) = 0$ 只有零特征值, 故 $\varphi(\lambda) = \lambda^2 - 3\lambda + 2 = 0$, 于是 $\lambda = 1$ 或 2.

10. 设 A 为正交阵, 且 $|A| = -1$, 证明 $\lambda = -1$ 是 A 的特征值.

证 $\because |A+E| = |A+AA^{-1}| = |A(E+A^{-1})| = |A||E+A^{-1}| = -|E+A^T| = -|A+E|$

$\therefore |A+E| = 0$, 故 $\lambda = -1$ 是 A 的特征值.

11. 设 $\lambda \neq 0$ 是 m 阶矩阵 $A_{m \times m} B_{n \times m}$ 的特征值, 证明 λ 也是 n 阶矩阵 $B_{n \times m} A_{m \times m}$ 的特征值

证 $\because \lambda \neq 0$ 是 m 阶矩阵 $A_{m \times m} B_{n \times m}$ 的特征值, $\therefore \exists p \neq 0$, 使得 $(AB)p = \lambda p$.

而 $Bp \neq 0$, 否则 $(AB)p = A(Bp) = 0$,

而 $\lambda p \neq 0$ 矛盾. 又 $B(AB)p = B(\lambda p)$, 即 $(BA)(Bp) = \lambda(Bp)$, $\therefore \lambda$ 也是 n 阶矩阵 $B_{n \times m} A_{m \times m}$ 的特征值

12. 已知 3 阶矩阵 A 的特征值为 1, 2, 3, 求 $|A^3 - 5A^2 + 7A|$.

解: 记 $\varphi(A) = A^3 - 5A^2 + 7A$, 若 λ 是 A 的特征值, 则 $\varphi(\lambda) = \lambda^3 - 5\lambda^2 + 7\lambda$ 是 $\varphi(A)$ 的特征值,

即 $\varphi(1) = 3, \varphi(2) = 2, \varphi(3) = 3$ 是 $A^3 - 5A^2 + 7A$ 的特征值, $\therefore |A^3 - 5A^2 + 7A| = 3 \times 2 \times 3 = 18$.

13. 已知 3 阶矩阵 A 的特征值为 1, 2, -3, 求 $|A' + 3A + 2E|$.

解: $\varphi(A) = A' + 3A + 2E = |A|A^{-1} + 3A + 2E = -6A^{-1} + 3A + 2E$,

有 $\varphi(\lambda) = -6\lambda^{-1} + 3\lambda + 2$ 是 $\varphi(A)$ 的特征值, $\varphi(1) = -1, \varphi(2) = 5, \varphi(-3) = -5$,

$\therefore |A' + 3A + 2E| = -1 \times 5 \times (-5) = 25$.

14. 设 A, B 都是 n 阶矩阵, 且 A 可逆, 证明 AB 与 BA 相似.

证 $\because A$ 可逆, 即 A^{-1} 存在, 而 $A^{-1}(AB)A = BA$, $\therefore AB$ 与 BA 相似.

15. 设 $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & x \\ 4 & 0 & 5 \end{pmatrix}$ 可相似对角化, 求 x .

证 $|A - \lambda E| = \begin{vmatrix} 2-\lambda & 0 & 1 \\ 3 & 1-\lambda & x \\ 4 & 0 & 5-\lambda \end{vmatrix} = (1-\lambda)^2(6-\lambda)$, 得 $\lambda_1 = 6, \lambda_2 = \lambda_3 = 1$.

由 $A - 6E = \begin{pmatrix} -4 & 0 & 1 \\ 3 & -5 & x \\ 4 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} -4 & 0 & 1 \\ 0 & -20 & 4x+3 \\ 0 & 0 & 0 \end{pmatrix}$, 得相应于特征值 $\lambda_1 = 6$ 的线性无关的特征向量为一个,

故 A 可对角化的充要条件是相应于特征值 $\lambda_2 = \lambda_3 = 1$ 的线性无关的特征向量为两个, 则 $R(A - E) = 1$,

由 $A - E = \begin{pmatrix} 1 & 0 & 1 \\ 3 & 0 & x \\ 4 & 0 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & x-3 \\ 0 & 0 & 0 \end{pmatrix}$ 得 $x-3=0$, 即 $x=3$. 因此, $x=3$ 时 A 可相似对角化.

16. 已知 $p = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ 是矩阵 $A = \begin{pmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{pmatrix}$ 的一个特征向量.

(1) 求参数 a, b 及特征向量 p 所对应的特征值;

(2) 问 A 能不能相似对角化? 并说明理由.

解(1) $Ap = \lambda p \Rightarrow \begin{pmatrix} -1 \\ 2+a \\ 1+b \end{pmatrix} = \begin{pmatrix} \lambda \\ \lambda \\ -\lambda \end{pmatrix} \Rightarrow \lambda = -1, a = -3, b = 0;$

(2) $|A - \lambda E| = \begin{vmatrix} 2-\lambda & -1 & 2 \\ 5 & -3-\lambda & 3 \\ -1 & 0 & -2-\lambda \end{vmatrix} = -(1+\lambda)^3$, A 的特征值为 $\lambda_1 = \lambda_2 = \lambda_3 = -1$

$A - (-1)E = \begin{pmatrix} 3 & -1 & 2 \\ 5 & -2 & 3 \\ -1 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, 故 A 的线性无关的特征向量个数为 $1 < 3$,

所以 A 不能相似对角化

17. 设 $A = \begin{pmatrix} 1 & 4 & 2 \\ 0 & -3 & 4 \\ 0 & 4 & 3 \end{pmatrix}$, 求 A^{100} .

解: A 的特征多项式为 $|A - \lambda E| = \begin{vmatrix} 1-\lambda & 4 & 2 \\ 0 & -3-\lambda & 4 \\ 0 & 4 & 3-\lambda \end{vmatrix} = (1-\lambda)(\lambda-5)(\lambda+5),$

$\therefore A$ 的特征值为 $\lambda_1 = 1, \lambda_2 = 5, \lambda_3 = -5$

$\therefore A - E = \begin{pmatrix} 0 & 4 & 2 \\ 0 & -4 & 4 \\ 0 & 4 & 2 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$

$\therefore$ 相应于特征值 $\lambda_1 = 1$ 的特征向量 $p_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

18. 在某国, 每年有比例为 p 的农村居民移居城镇, 有比例为 q 的城镇居民移居农村. 假设该国总人口数不变, 且上述人口迁移的规律也不变. 把 n 年后农村人口和城镇人口占总人口的比例依次记为 x_n 和 y_n ($x_n + y_n = 1$).

(1) 求关系式 $\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = A \begin{pmatrix} x_n \\ y_n \end{pmatrix}$ 中的矩阵 A ;

(2) 设目前农村人口与城镇人口相等, 即 $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$, 求 $\begin{pmatrix} x_n \\ y_n \end{pmatrix}$.

解(1) 由题意知 $x_{n+1} = (1-p)x_n + qy_n, y_{n+1} = (1-q)y_n + px_n$, 故 $A = \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix}.$

(2) A 的特征多项式为 $|A - \lambda E| = \begin{vmatrix} 1-p-\lambda & q \\ p & 1-q-\lambda \end{vmatrix} = (\lambda-1)(\lambda-1+p+q),$

故特征值为 $\lambda_1 = 1, \lambda_2 = 1-p-q$

$A - E = \begin{pmatrix} -p & q \\ p & -q \end{pmatrix} \sim \begin{pmatrix} -p & q \\ 0 & 0 \end{pmatrix}$, 相应于特征值 $\lambda_1 = 1$ 的特征向量为 $\begin{pmatrix} q \\ p \end{pmatrix},$

$A - (1-p-q)E = \begin{pmatrix} q & q \\ p & p \end{pmatrix} \sim \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, 相应于特征值 $\lambda_2 = 1-p-q$ 的特征向量为 $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

令 $P = \begin{pmatrix} q & 1 \\ p & -1 \end{pmatrix}, P^{-1} = \frac{1}{p+q} \begin{pmatrix} 1 & 1 \\ p & -q \end{pmatrix}, P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & 1-p-q \end{pmatrix}$

$A^n = P \begin{pmatrix} 1 & 0 \\ 0 & (1-p-q)^n \end{pmatrix} P^{-1} = \frac{1}{p+q} \begin{pmatrix} q+pr^n & q-qr^n \\ p-pr^n & p+qr^n \end{pmatrix}$

19. 试求一个正交的相似变换矩阵, 将下列对陈阵化为对角阵:

$$(1) \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}; (2) \begin{pmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}.$$

解: (1) $|A - \lambda E| = \begin{vmatrix} 2-\lambda & -2 & 0 \\ -2 & 1-\lambda & -2 \\ 0 & -2 & -\lambda \end{vmatrix} = (1-\lambda)(\lambda-4)(\lambda+2)$, 故得特征值为 $\lambda_1 = -2$ $\lambda_2 = 1$ $\lambda_3 = 4$

当 $\lambda_1 = -2$ 时, $A + 2E = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & -2 \\ 0 & -2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$,

线性无关的特征向量可取为 $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, 单位化得 $p_1 = \begin{pmatrix} 1/\sqrt{3} \\ 2/\sqrt{3} \\ 2/\sqrt{3} \end{pmatrix}$

当 $\lambda_2 = 1$ 时, $A - E = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 0 & -2 \\ 0 & -2 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{pmatrix}$,

线性无关的特征向量可取为 $\begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$, 单位化得 $p_2 = \begin{pmatrix} -2/\sqrt{3} \\ -1/\sqrt{3} \\ 2/\sqrt{3} \end{pmatrix}$

当 $\lambda_3 = 4$ 时, $A - 4E = \begin{pmatrix} -2 & -2 & 0 \\ -2 & -3 & -2 \\ 0 & -2 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$,

线性无关的特征向量可取为 $\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$, 单位化得 $p_3 = \begin{pmatrix} 2/\sqrt{3} \\ -2/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$

得正交阵 $P = (p_1, p_2, p_3) = \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{3} & 2/\sqrt{3} \\ 2/\sqrt{3} & -1/\sqrt{3} & -2/\sqrt{3} \\ 2/\sqrt{3} & 2/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$,

$P^{-1}AP = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

(2) $|A - \lambda E| = \begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & 5-\lambda & -4 \\ -2 & -4 & 5-\lambda \end{vmatrix} = -(\lambda-1)^2(\lambda-10)$, 故得特征值为 $\lambda_1 = \lambda_2 = 1$ $\lambda_3 = 10$

当 $\lambda_1 = \lambda_2 = 1$ 时, 由 $\begin{pmatrix} 1 & 2 & -2 \\ 2 & 4 & -4 \\ -2 & -4 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 线性无关的特征向量可取为 $\xi_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ $\xi_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$

正交化得 $\xi_1' = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ $\xi_2' = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{4}{5} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4/5 \\ 1 \\ 1 \end{pmatrix}$, 单位化得 $p_1 = \begin{pmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{pmatrix}$ $p_2 = \begin{pmatrix} 2\sqrt{5}/15 \\ 4\sqrt{5}/15 \\ \sqrt{5}/3 \end{pmatrix}$

当 $\lambda_3 = 10$ 时, 由 $\begin{pmatrix} -8 & 2 & -2 \\ 2 & -5 & -4 \\ -2 & -4 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, 线性无关的特征向量可取为 $\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$, 单位化得 $p_3 = \begin{pmatrix} -1/\sqrt{3} \\ -2/\sqrt{3} \\ 2/\sqrt{3} \end{pmatrix}$

于是得正交阵 $P = (p_1, p_2, p_3) = \begin{pmatrix} -2/\sqrt{5} & 2\sqrt{5}/15 & -1/\sqrt{3} \\ 1/\sqrt{5} & 4\sqrt{5}/15 & -2/\sqrt{3} \\ 0 & \sqrt{5}/3 & 2/\sqrt{3} \end{pmatrix}$, $P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10 \end{pmatrix}$

20. 设矩阵 $A = \begin{pmatrix} 1 & -2 & -4 \\ -2 & x & -2 \\ -4 & -2 & 1 \end{pmatrix}$ 与 $\Lambda = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & y \end{pmatrix}$ 相似, 求 x, y ; 并求一个正交阵 P , 使 $P^{-1}AP = \Lambda$.

解 方阵 A 与 Λ 相似, 则 A 与 Λ 特征多项式相同, 即

$$|A - \lambda E| = |\Lambda - \lambda E| \Rightarrow \begin{vmatrix} 1-\lambda & -2 & -4 \\ -2 & x-\lambda & -2 \\ -4 & -2 & 1-\lambda \end{vmatrix} = \begin{vmatrix} 5-\lambda & 0 & 0 \\ 0 & -4-\lambda & 0 \\ 0 & 0 & y-\lambda \end{vmatrix} \Rightarrow \begin{cases} x=4 \\ y=5 \end{cases}$$

A 的特征值为 $\lambda_1 = \lambda_2 = 5$ $\lambda_3 = -4$

当 $\lambda_1 = \lambda_2 = 5$ 时, 由 $A - 5E = \begin{pmatrix} -4 & -2 & -4 \\ -2 & -1 & -2 \\ -4 & -2 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1/2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 正交的特征向量可取为 $\xi_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\xi_2 = \begin{pmatrix} 1 \\ -4 \\ 1 \end{pmatrix}$

单位化得 $p_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, $p_2 = \begin{pmatrix} \frac{1}{3\sqrt{2}} \\ \frac{-4}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \end{pmatrix}$

当 $\lambda_3 = -4$ 时, 由 $A + 4E = \begin{pmatrix} 5 & -2 & -4 \\ -2 & 8 & -2 \\ -4 & -2 & 5 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}$, 特征向量可取为 $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$, 单位化得 $p_3 = \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$

于是令 $P = (p_1, p_2, p_3) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} & \frac{1}{3\sqrt{2}} \\ 0 & \frac{1}{3} & \frac{-4}{3\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{2}{3} & \frac{1}{3\sqrt{2}} \end{pmatrix}$, 有 $P^{-1}AP = \Lambda$.

21. 设3阶矩阵 A 的特征值为 $\lambda_1 = 2, \lambda_2 = -2, \lambda_3 = 1$, 对应的特征向量依次为 $p_1 = (0, 1, 1)^T, p_2 = (1, 1, 1)^T, p_3 = (1, 1, 0)^T$ 求 A .

解: 记 $P = (p_1, p_2, p_3) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, 则 $P^{-1} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix}$, 于是 $A = P \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix} P^{-1} = \begin{pmatrix} -2 & 3 & -3 \\ -4 & 5 & -3 \\ -4 & 4 & -2 \end{pmatrix}$

22. 设3阶矩阵 A 的特征值为 $\lambda_1 = 1, \lambda_2 = -1, \lambda_3 = 0$, 对应的特征向量依次为 $p_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, p_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$, 求 A .

解: 设相应于 λ_i 的特征向量为 p_i , 由于对称阵的属于不同特征值的特征向量正交, 则

$$p_1^T p_3 = 0, p_2^T p_3 = 0 \Rightarrow \begin{cases} x_1 + 2x_2 + 2x_3 = 0 \\ 2x_1 + x_2 - 2x_3 = 0 \end{cases} \Rightarrow p_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

将 p_1, p_2, p_3 单位化, 得 $Q_1 = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, Q_2 = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, Q_3 = \frac{1}{3} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$,

记 $Q = (Q_1, Q_2, Q_3) = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$, 则 $Q^T A Q = \Lambda = \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix}$

于是 $A = Q \Lambda Q^T = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 & 0 & 2 \\ 0 & 1 & 2 \\ 2 & 2 & 0 \end{pmatrix}$

23. 设3阶对称阵 A 的特征值为 $\lambda_1 = 6, \lambda_2 = \lambda_3 = 3$, 与特征值 $\lambda_1 = 6$ 对应的特征向量为 $p_1 = (1, 1, 1)^T$, 求 A .

解: 设 $A = \begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_4 & x_5 \\ x_3 & x_5 & x_6 \end{pmatrix}$, 由 $A p_1 = 6 p_1$, 得 $\begin{cases} x_1 + x_2 + x_3 = 6 \\ x_2 + x_4 + x_5 = 6 \\ x_3 + x_5 + x_6 = 6 \end{cases} \quad (1),$

$\therefore 3$ 是对称阵 A 的2重特征值, $\therefore R(A - 3E) = 1$,

而 $A - 3E = \begin{pmatrix} x_1 - 3 & x_2 & x_3 \\ x_2 & x_4 - 3 & x_5 \\ x_3 & x_5 & x_6 - 3 \end{pmatrix} \sim_{(r_1+r_2+r_3)/3} \begin{pmatrix} 1 & 1 & 1 \\ x_2 & x_4 - 3 & x_5 \\ x_3 & x_5 & x_6 - 3 \end{pmatrix}$

得 $\begin{cases} x_2 = x_4 - 3 = x_5 \\ x_3 = x_5 = x_6 - 3 \end{cases} \quad (2),$ 由(1)(2)联立解得 $x_1 = x_4 = x_6 = 4, x_2 = x_3 = x_5 = 1$, 即 $A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix}$.

24. 设 $a = (a_1, a_2, \dots, a_n)^T, a_1 \neq 0, A = aa^T$,

(1) 证明 $\lambda = 0$ 是 A 的 $n-1$ 重特征值; (2) 求 A 的非零特征值及 n 个线性无关的特征向量.

$$(1) \text{ 证 } \because A = aa^T = \begin{pmatrix} a_1^2 & a_1 a_2 & \cdots & a_1 a_n \\ a_2 a_1 & a_2^2 & \cdots & a_2 a_n \\ \cdots & \cdots & \cdots & \cdots \\ a_n a_1 & a_n a_2 & \cdots & a_n^2 \end{pmatrix} \sim \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 0 \end{pmatrix},$$

$\therefore R(A) = 1$, 则 $|A - 0E| = 0$, 又 $R(A - 0E) = 1 = n - (n-1)$, $\therefore \lambda = 0$ 是 A 的 $n-1$ 重特征值;

(2) 解 记 A 的 n 个特征值为 $\lambda_1 = \lambda_2 = \cdots = \lambda_{n-1} = 0, \lambda_n \neq 0$,

$$\because \lambda_1 + \lambda_2 + \cdots + \lambda_{n-1} + \lambda_n = a_1^2 + a_2^2 + \cdots + a_n^2, \therefore \lambda_n = a_1^2 + a_2^2 + \cdots + a_n^2.$$

25. (1) 设 $A = \begin{pmatrix} 3 & -2 \\ -2 & 3 \end{pmatrix}$, 求 $\varphi(A) = A^{10} - 5A^9$;

(2) 设 $A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}$, 求 $\varphi(A) = A^{10} - 6A^9 + 5A^8$.

解: (1) $|A - \lambda E| = \begin{vmatrix} 3-\lambda & -2 \\ -2 & 3-\lambda \end{vmatrix} = (1-\lambda)(5-\lambda)$, 得 $\lambda_1 = 1, \lambda_2 = 5$.

$$A - E = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \text{ 得 } p_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, A - 5E = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \text{ 得 } p_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\text{令 } P = (p_1, p_2) = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, P^{-1}AP = \Lambda = \begin{pmatrix} 1 & \\ & 5 \end{pmatrix}, A = P\Lambda P^{-1},$$

$$\varphi(A) = P\varphi(\Lambda)P^{-1} = P \begin{pmatrix} \varphi(1) & \\ & \varphi(5) \end{pmatrix} P^{-1} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -4 & \\ & 0 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix}$$

$$(2) |A - \lambda E| = \begin{vmatrix} 2-\lambda & 1 & 2 \\ 1 & 2-\lambda & 2 \\ 2 & 2 & 1-\lambda \end{vmatrix} = -(\lambda-1)(\lambda+1)(\lambda-5), \text{ 得 } \lambda_1 = 1, \lambda_2 = 5, \lambda_3 = -1.$$

$$A - E = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得 } p_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},$$

$$A - 5E = \begin{pmatrix} -3 & 1 & 2 \\ 1 & -3 & 2 \\ 2 & 2 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得 } p_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$A + E = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得 } p_3 = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$$

$$\text{令 } P = (p_1, p_2, p_3) = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ 0 & 1 & 2 \end{pmatrix}, P^{-1} = \frac{1}{6} \begin{pmatrix} -3 & 3 & 0 \\ 2 & 2 & 2 \\ -1 & -1 & 2 \end{pmatrix}, \Lambda = \begin{pmatrix} 1 & & \\ & -1 & \\ & & 5 \end{pmatrix}$$

$$A = P\Lambda P^{-1}, \varphi(A) = P\varphi(\Lambda)P^{-1} = P \begin{pmatrix} \varphi(\lambda_1) & & \\ & \varphi(\lambda_2) & \\ & & \varphi(\lambda_3) \end{pmatrix} P^{-1}$$

26. 用矩阵记号表示下列二次型:

$$(1) f = x^2 + 4xy + 4y^2 + 2xz + z^2 + 4yz;$$

$$(2) f = x^2 + y^2 - 7z^2 - 2xy - 4xz - 4yz;$$

$$(3) f = x_1^2 + x_2^2 + x_3^2 + x_4^2 - 2x_1x_2 + 4x_1x_3 - 2x_1x_4 + 6x_2x_3 - 4x_2x_4.$$

$$\text{解: (1) } f = (x, y, z) \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$(2) f = (x, y, z) \begin{pmatrix} 1 & -1 & -2 \\ -1 & 1 & -2 \\ -2 & -2 & -7 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$(3) f = (x_1, x_2, x_3, x_4) \begin{pmatrix} 1 & -1 & 2 & -1 \\ -1 & 1 & 3 & -2 \\ 2 & 3 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

27. 写出下列二次型的矩阵:

$$(1) f(x) = x^T \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} x; \quad (2) f(x) = x^T \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} x$$

$$\text{解: (1) } \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix} \quad (2) \begin{pmatrix} 1 & 3 & 5 \\ 3 & 5 & 7 \\ 5 & 7 & 9 \end{pmatrix}$$

28. 求一个正交变换使下列二次型成标准型:

$$(1) f = 2x_1^2 + 3x_2^2 + 3x_3^2 + 4x_2x_3;$$

$$(2) f = x_1^2 + x_2^2 + 2x_1x_2 - 2x_2x_3.$$

$$\text{解: (1) 二次型的矩阵为 } A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 2 \\ 0 & 2 & 3 \end{pmatrix}$$

$$|A - \lambda E| = \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 2 \\ 0 & 2 & 3-\lambda \end{vmatrix} = (2-\lambda)(5-\lambda)(1-\lambda), \text{ 故 } \lambda \text{ 的特征值为 } \lambda_1 = 2, \lambda_2 = 5, \lambda_3 = 1.$$

$$\text{当 } \lambda_1 = 2 \text{ 时, 解方程 } (A - 2E)x = 0, \text{ 由 } A - 2E = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } p_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{当 } \lambda_2 = 5 \text{ 时, 解方程 } (A - 5E)x = 0, \text{ 由 } A - 5E = \begin{pmatrix} -3 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } p_2 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\text{当 } \lambda_3 = 1 \text{ 时, 解方程 } (A - E)x = 0, \text{ 由 } A - E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } p_3 = \begin{pmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\text{于是正交变换为 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}, \text{ 且有 } f = 2y_1^2 + 5y_2^2 + y_3^2$$

$$(2) \text{ 二次型的矩阵为 } A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{pmatrix},$$

$$|A - \lambda E| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 1 & -\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = (2-\lambda)(\lambda-1)(\lambda+1), \text{ 特征值为 } \lambda_1 = 2, \lambda_2 = 1, \lambda_3 = -1.$$

$$\text{当 } \lambda_1 = 2 \text{ 时, 解方程 } (A - 2E)x = 0, \text{ 由 } A - 2E = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } p_1 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\text{当 } \lambda_2 = 1 \text{ 时, 解方程 } (A - E)x = 0, \text{ 由 } A - E = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } p_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

当 $\lambda_3 = -1$ 时, 解方程 $(A+E)x = 0$, 由 $A+E = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$ 得 $p_3 = \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$

于是正交变换为 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, 且有 $f = 2y_1^2 + y_2^2 - y_3^2$

29. 求一个正交变换把二次曲面的方程 $3x^2 + 5y^2 + 5z^2 + 4xy - 4xz - 10yz = 1$ 化成标准方程.

解: 二次型 $f = 3x^2 + 5y^2 + 5z^2 + 4xy - 4xz - 10yz$ 所对应的矩阵为 $A = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 5 & -5 \\ -2 & -5 & 5 \end{pmatrix}$

$|A - \lambda E| = \begin{vmatrix} 3-\lambda & 2 & -2 \\ 2 & 5-\lambda & -5 \\ -2 & -5 & 5-\lambda \end{vmatrix} = (11-\lambda)(\lambda-2)\lambda$, 特征值为 $\lambda_1 = 11, \lambda_2 = 2, \lambda_3 = 0$

相应于 $\lambda_1 = 11$, 由 $A - 11E = \begin{pmatrix} -8 & 2 & -2 \\ 2 & -6 & -5 \\ -2 & -5 & -6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, 得 $p_1 = \begin{pmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix}$

相应于 $\lambda_2 = 2$, 由 $A - 2E = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 3 & -5 \\ -2 & -5 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, 得 $p_2 = \begin{pmatrix} -\frac{4}{3\sqrt{2}} \\ -\frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \end{pmatrix}$

相应于 $\lambda_3 = 0$, 由 $A - 0E = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 5 & -5 \\ -2 & -5 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$, 得 $p_3 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

于是, 正交变换为 $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} & -\frac{4}{3\sqrt{2}} & 0 \\ \frac{2}{3} & -\frac{1}{3\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{2}{3} & \frac{1}{3\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}$, 曲面方程为 $1u^2 + 2v^2 = 1$